Seal Team 2 MTD Rebuttal Document

Commentary: A significant portion of our Midterm Design Document was rewritten due to the quantitization of Dr. Van Neste's specifications and constraints, and the establishment of the DAQ subsystem. Therefore, not all changes that were made to the document can be mapped directly to the specified critiques given to us in the graded rubric. Because of this, we have provided below all changes that apply to the graded rubric critiques, but also encourage you to reread the entire document to understand how the changes we have made our paper into a more cohesive document.

- Seal Team 2

Project Proposal Revisions:

1.

Critique:

P3. Problem Statement Clear, concise, and covers all essential aspects?	**One to three sentences naming the need, the user, the context, and a measurable success condition. Solution-neutral.**	**Clear and complete. Slightly long or hints at a solution.**	Understandable but missing one element, or states a solution instead of a need ("Design an MPPT charger").	Vague ("build a better charger"), or several conflicting statements.	No problem statement.	**4.5**

Instructor: Complete scope, but much longer than the rubric target and it embeds selected implementation choices.

Previous Problem statement:

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**Seal Team 2** shall design a Telluric Current Sensor that obtains reliable
current density information of the telluric currents emmitted from a substation
while adhering to the **TCS-DS** specifications and DAQ subsystem constraints.
The Sensor shall contain an MCU that captures the telluric current data
digitally, extracts the 60 Hz, 180 Hz, and 540 Hz amplitudes from an FFT
algorithm, determines the existence of a ground fault, and sends the data
serially to a telecommunications module while adhering to the **TCS-MS**
specifications and MCU subsystem constraints. The Sensor shall contain a
telecommunications module to transmit the data across a distance of 1 km to a
gateway accessible by all stakeholders while adhering to the **TCS-TS**
specifications and telecommunication subsystem constraints. The Sensor shall be
powered locally with minimal maintenance needs as per the **TCS-PS**
specifications and power subsystem constraints. Finally, the sensor shall be
housed in a waterproof and weatherproof encasement, and should be capable of
operating in all outdoor conditions as per the **TCS-WS** specifications and
weatherization subsystem constraints.
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New Problem Statement:

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### Fully Formulated Problem Statement

Seal Team 2 shall design a Telluric Current Sensor capable of reliably
measuring telluric current density emitted from a substation, processing the
measurements to identify potential ground faults, and wirelessly transmitting
the resulting data to a gateway accessible to all stakeholders over a distance
of up to 1 km. The system shall operate autonomously with minimal maintenance
requirements and withstand outdoor environmental conditions while satisfying
all TCS-DS, TCS-MS, TCS-TS, TCS-PS, and TCS-WS specifications and subsystem
constraints.
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Fixes made: The problem statement keeps the same, complete scope as desired, but is now within the stated requirements of being 1-3 sentences. The problem statement is now solution neutral.

2.

· Critique:

Instructor: Complete scope, but much longer than the rubric target and it embeds selected implementation choices.						
P4. Stakeholder Specifications Listed specifications and measures of success?	Every specification is quantitative with units, tolerance, and a verification method, and traces to a stakeholder need.	Quantitative with units. Traceability or verification missing for a few.	**Mix of quantitative and qualitative terms ("fast", "efficient").**	Mostly qualitative ("high efficiency, safe, low cost"). No measures of success.	No specifications listed.	**3**
Instructor: Many specifications are vague or untestable: "minimized," "weatherproof," "all outdoor conditions," "abnormalities," and "efficiently." The 24-hour power target conflicts with battery replacement >1 month.						

All of the stakeholder specifications were turned into completely measurable quantitative specifications. Each of the changes are shown below. To see the changes, please compare the Specifications and Constraints subsection of project proposal between the first submission and the newest commit. Experiments were also created for every specification and enumerated in a table in the Measures of Success subsection of project proposal, to prove measurability and provide a verification procedure.

· Text Before Revision:

Specifications

The following section will enumerate and clarify the specifications provided by the customer, Dr. Van Neste, on behalf of all of the stakeholders (See Stakeholders subsection)

Telecommunications Subsystem Specifications

- **TCS-TS1** The customer, Dr. Charles Van Neste, specifies that the data must be transmitted wirelessly to a gateway accesible by any of the stakeholders that is at most 1 km away from the sensor.
- **TCS-TS2** The customer, Dr. Charles Van Neste, specifies that the power consumption of the transmitter must be minimized.
- **TCS-TS3** The customer, Dr. Charles Van Neste, specifies that the entire communication module must be weatherized.

MCU Subsystem Specifications

- **TCS-MS1** The customer, Dr. Charles Van Neste, specifies that the Telluric Current Sensor may use the Particle Boron Microcontroller as the combined MCU and Telecommunication modules.
- **TCS-MS2** The customer, Dr. Charles Van Neste, specifies that the MCU must use a Fast Fourier Transform algorithm to extract the 60 Hz, 180 Hz, and 540 Hz frequency amplitudes from the sampled telluric current input.
- **TCS-MS3** The customer, Dr. Charles Van Neste, specifies that the MCU must detect abnormalities in the frequencies and determine if a fault has occured based on such data.

Weatherization Subsystem Specifications

- **TCS-WS1** The customer, Dr. Charles Van Neste, specifies that the entire system must be housed in a water proof and weather proof encasement.
- **TCS-WS2** The customer, Dr. Charles Van Neste, specifies that all components must be capable of operating under all of Cookeville, TN temperature ranges ($-30 - 41^{\circ}$ C).
- **TCS-WS3** The customer, Dr. Charles Van Neste, specifies that the system must be able to operate under the extreme weather conditions in Cookeville: rain, snow, etc.

DAQ Subsystem Specifications

- **TCS-DS1** The customer, Dr. Charles Van Neste, specifies that the system must measure the voltage differences between two or more electrodes and input this data into the MCU via an ADC converter.

Power Subsystem Specifications

- **TCS-PS1** The customer, Dr. Charles Van Neste, specifies that the system must be powered locallly and efficiently and require minimal maintenance. (Battery replacements >1 month).
- **TCS-PS2** The customer, Dr. Charles Van Neste, specifies that the system must incorporate surge protection devices complying with IEEE Std C62.41 to withstand high-voltage transient overvoltages a lightning-induced surges.
- **TCS-PS3** The customer, Dr. Charles Van Neste, specifies that the system may be charged via the telluric currents from the substation through the use of a supercapacitor bank, thereby requiring no charging element.

PCB Specifications

- **TCS-PCB1** The customer, Dr. Charles Van Neste, specifies that each of the subsystems may be connected via a printed circuit board to make the commercialization of this project more effective.

Discussion

Upon negotiation with Dr. Charles Van Neste and extensive review of the TCS-MS1 specification, **Seal Team 2** decided to void TCS-MS1 in lieu of better options for microcontroller units that will be outlined in the Conceptual Design Document. Furthermore, due to the complexity of the supercapacitor bank mentioned in **TCS-PS3**, **Seal Team 2** negotiated with the customer to void this requirement. All other specifications will be considered for the design of the Telluric Current Sensor.

- Text Post Revision:

Specifications

The following section will enumerate and clarify the specifications provided by the customer, Dr. Van Neste, on behalf of all of the stakeholders (See Stakeholders subsection). All of the following specifications will be binary. they are either achieved or failed with quantitative boundaries so as to dilineate success from failure clearly. The tests which evaluate whether each subsystem has passed or failed the specifications will be identified later in *Measures of Success*.

Telecommunications Subsystem Specifications

- **TCS-TS1:** The customer, Dr. Van Neste specifies that the telluric current sensor shall wirelessly transmit measurement data to a designated reciever located up to 1,000 m from the sensor, measured as straight-line distance from transmitter to reciever.
- **TCS-TS2:** The communication link shall successfully deliver at least 99% of scheduled data transmissions under all deployment conditions (as per **TCS-WS2** and **TCS-WS3**).
- **TCS-TS3:** The gateway shall make received measurement data accessible, within 5 minutes of reception, to all designated project stakeholders through an authenticated interface.
- **TCS-TS4:** The authenticated interface shall store all historical measurement data within a 30 day period.

MCU Subsystem Specifications

- **TCS-MS1** The customer, Dr. Charles Van Neste, specifies that the Telluric Current Sensor may use the Particle Boron Microcontroller as the combined MCU and Telecommunication modules.
- **TCS-MS2** The customer, Dr. Charles Van Neste, specifies that the MCU must use a Fast Fourier Transform algorithm to extract the frequency amplitudes of the first twenty 60 Hz harmonics of the measured telluric currents
- **TCS-MS2.1** The sampling rate of the measured data should be at least five times the twentieth harmonic of 60Hz (i.e >6000 Hz) to mitigate aliasing.
- **TCS-MS3** Dr. Charles Van Neste specifies that the MCU shall send the following data per the following procedures to the Telecommunication Subsystem:

- **TCS-MS3.1** During regular operation (i.e. no fault detected), the MCU shall send a transmission packet every hour containing the 20 FFT extracted amplitudes calculated 60 minutes prior, 55 minutes prior,..., and 5 minutes prior to transmission. The MCU shall also send a boolean indicator value of FALSE indicating the lack of a fault.
- **TCS-MS3.2** Within the first 10 minutes of detection of an abnormality in the measurement data*, the MCU shall send a priority transmission packet to the telecommunications unit containing the 20 FFT extracted amplitudes between the time of detection and 60 seconds after detection sampled at a rate of at least 100 samples per second. The MCU shall also send a boolean indicator value of TRUE indicating the presence of a fault.

Weatherization Subsystem Specifications

- **TCS-WS1** The customer, Dr. Charles Van Neste, specifies that the entire system must be housed in a waterproof* and weatherproof encasement.
- **TCS-WS2** The customer, Dr. Charles Van Neste, specifies that all components must be capable of operating under all of Cookeville, TN temperature ranges (-30 to 41° C).

  *Waterproof equates that the system's sensitive electrical equipment will remain operational after the enclosure has been submerged in water for 60 minutes.

DAQ Subsystem Specifications

- **TCS-DS1** The customer, Dr. Charles Van Neste, specifies that the DAQ subsystem must measure the voltage differences between two or more electrodes and input this data into the MCU via an ADC converter.
  - **TCS-DS1.1** This data must be within 10% of the true RMS voltage of the telluric current waveforms as measured by a commercial multimeter whenever the true RMS voltage is above 1 mV and below 10 V.
  - **TCS-DS1** This data must be within 20% of the true RMS voltage of the telluric current waveforms as measured by a commercial multimeter whenever the true RMS voltage is at or above 10 V and less than 500 V.

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    - **TCS-DS1** This data must be within 20% of the true RMS voltage of the telluric current waveforms as measured by a commercial multimeter whenever the true RMS voltage is at or above 10 V and less than 500 V.
- **TCS-DS2** The customer, Charles Van Neste, specifies that the DAQ subsystem must be compatible with and within the input range of the MCU in every operation case aside from ultra-high voltage surges of 500 V or greater.

Power Subsystem Specifications

- **TCS-PS1** The customer, Dr. Charles Van Neste, specifies that the system must be powered locally and require power system maintenace to be done as infrequently as possible (4 battery changes a year at maximum).
- **TCS-PS2** The customer, Dr. Charles Van Neste, specifies that the system must incorporate surge protection devices complying with IEEE Std C62.41 to withstand high-voltage transient overvoltages a lightning-induced surges.
- **TCS-PS3** The customer, Dr. Charles Van Neste, specifies that the system may be charged via the telluric currents from the substation through the use of a supercapacitor bank, thereby requiring no charging element.
- **TCS-PS4** The customer, Dr. Charles Van Neste, specifies that the output of the regulators and LDOs meet low-noise requirements, ensuring no measurable power supply leaks into the sensor's bandwidth.

Financial Specifications

- **TCS-F1** The entire system shall be implemented within the Capstone Design budget of $1000.00.
- **TCS-F2** The system shall cost at most $100 per year for constant operation.
    - **TCS-F2.1** The total power dissipated by the sensor should be minimized such that the total cost per year for power system maintenance does not exceed $85/year.
    - **TCS-F2.2** The total cost for the connectivity between the transmitter and reciever of the Telecommunications Subsystem should not exceed $15/year

PCB Specifications

- **TCS-PCB1** The customer, Dr. Charles Van Neste, specifies that each of the subsystems (excluding any surge protection or distant sensors) must be connected via a printed circuit board.

Discussion

Upon negotiation with Dr. Charles Van Neste and extensive review of the TCS-MS1 specification, **Seal Team 2** decided to void TCS-MS1 in lieu of better options for microcontroller units that will be outlined in the Conceptual Design Document. Furthermore, due to the complexity beyond the scope of the capstone course of the implementation of a supercapacitor bank mentioned in **TCS-PS3**, **Seal Team 2** negotiated with the customer to void this requirement. All other specifications will be considered for the design of the Telluric Current Sensor.

Measures of Success

Specifications and constraints are given to **Seal Team 2** which are specific for each subsystem, with a choice few applying to multiple or all subsystems. This section will be separated by subsystem and will outline how **Seal Team 2** will measure the success of each subsystem. As written in *Specifications and Constraints*, success is binary. Each of the following tests will be either passed or failed.

The following general structure will be a restatement of the specification or constraint as aforementioned with a binary test that will show success or failure. The reader should be viligant: the subsections do not list *all* of the specifications that apply to the subsystem rather those which fall primarily into each category. In this way the measures of success will not be repeated for each subsystem; rather, each section will define how those that apply majorly to the subsystem will be measured for the entre system.

In general, there is a budget for the entire system as imposed by **TCS-F1**, **TCS-F2**, **TCS-F2.1**, and **TCS-F2.2** and a wattage budget imposed by the cost of energy which applies to the aforementioned financial specifications. **TCS-F1** will be succeeded if **Seal Team 2** remains under budget. Specifications **TCS-F2**, **TCS-F2.1**, and **TCS-F2.2** will be succeeded if **Seal Team 2** remains under the wattage budget. Both of these budgets are defined more rigorously in *Budget*.

TCS-PCB1 is also golablly applicable to each of the systems (aside the physical protection). This specification will be accomplished so long as *all* of the specifications and constrains can be met wihle incorporating a PCB. TO further elaborate on the exclusion of the specification, any sensor that cannot be connected on the PCB in order to perform accurately will be left off. In addition to this, no surge protection nor anything unprotected from surges will be included on the PCB (fuses, TVS, GDT, etc.).

Data Acquisistion Measures of Success

The specifications majorly relevant to the DAQ are as follows:

- **TCS-DS1** The customer, Dr. Charles Van Neste, specifies that the DAQ subsystem must measure the voltage differences between two or more electrodes and input this data into the MCU via an ADC converter.
  - **TCS-DS1.1** This data must be within 10% of the true RMS voltage of the telluric current waveforms as measured by a commercial multimeter whenever the true RMS voltage is above 1 mV and below 10 V.
  - **TCS-DS1** This data must be within 20% of the true RMS voltage of the telluric current waveforms as measured by a commercial multimeter whenever the true RMS voltage is at or above 10 V and below 500 V.
- **TCS-DS2** The customer, Charles Van Neste, specifies that the DAQ subsystem must be compatible with and within the input range of the MCU in every operation case aside from ultra-high voltage surges of 500 V or greater.

TCS-DS1 will be succeded if **TCS-DS1.1** and **TCS-DS1.2** are met while incorporating an ADC and two or more electrodes.

TCS-DS1.1 will be succeeded if the following two tests pass:

Test 1: Apply a sine wave raging from 1 mV RMS to 10 V RMS to the electrodes of the DAQ system. For each sine wave, the digitally outputted sine wave must be within 10% accuracy of the applied sine wave. Alternatively it may be scaled by a scalar or added with a constant, but this scalar or constant but be constant with respect to the voltage. In this way, the input/output relationship will be confirmed in a controlled environment.

Test 2: Inject a current into the soil (using two electrodes to identify a path). Then, implant the sensor designed to measure the voltage between them. Measure the *same distance* using a multimeter and compare these values. As the last test states, it may be scaled by a scalar or added with a constant, but this scalar or constant but be constant with respect to the voltage. In this way, field accurate testing will be shown to be reliable.

For both of these tests, the RMS will be measured by DMM and compared against the RMS found by the following equation for calculating digital RMS.

$V_{RMS}=\sqrt{\frac{1}{n}\sum_{i=1}^nV_i^2}$

TCS-DS1.2 will be proven in the same way but with the different range as specified.

TCS-DS2 cannot be safely proven without sacrificing some equipment. The intent of ths specification is to have a reliable failsafe for the product. If by coincidence, lightning strikes the DAQ and it survives, this metric will be proven, but because we cannot conjure lightning, the success will rely on the datasheets of whatever surge protection devices are selected.

Telecommunications Measures of Success:

Specification/Constraint	Success Condition
TCS-TS1 The customer, Dr. Van Neste specifies that the telluric current sensor shall wirelessly transmit measurement data to a designated reciever located no more than 1,000 m from the sensor, measured as straight-line distance from transmitter to reciever.	Success of **TCS-TS1** is achieved if and only if the Telecommunications Subsystem can meet all of the requirements over a continuous 24 hour period for success of the **TCS-TS2**, **TCS-TS3**, and **TCS-TS4** specifications as shown below.
TCS-TS2: The communication link shall successfully deliver at least 99% of scheduled data transmissions under all deployment conditions (as per **TCS-WS2** and **TCS-WS3**).	Success of **TCS-TS2** is achieved if and only if the ratio of the total data sent to the transmitter over the total received data is at least 0.99 for all of three 24 hour test under the following conditions: - [24 hour Transmitter Test 1] The distance between transmitter and receiver is at least 1050 m measured as a straight line from transmitter to receiver, and a data packet is sent every hour from transmitter to receiver containing precisely the amount of information that is contained in the packet specified in **TCS-MS3.1.** - [24 hour Transmitter Test 2] The conditions of [24 hour Transmitter Test 1] are applied, but 6 additional priority transmissions are also chosen at random that each contain a packet with the same amount of information as the priority transmission packet specified in **TCS-MS3.2** - [24 hour Transmitter Test 3] The conditions of [24 hour Transmitter Test 2] are applied while the device is under the extreme temperature conditions of Cookeville TN (Specify).
TCS-TS3: The gateway shall make received measurement data accessible, within 5 minutes of reception, to all designated project stakeholders through an authenticated interface.	Success of the **TCS-TS3** specification will be acheived if and only if, under the same conditions of each of the three tests in **TCS-TS2**, the receiver sends all of the received data to an authenticated device, and the authenticated device generates a file that is accessible within 240 seconds of the reception containing at least 99% of the transmitted data

TCS-TS4: The authenticated interface shall store all historical measurement data within a 30 day period.	Success of the **TCS-TS4** specification will be acheived if and only if the authenticated data collection device passes the following test: [Authenticated Device Storage Test] A random set of data containing precisely 150 times the amount of bits sent across the transmitter over the [24 hour Transmitter Test 2] is to be sent to the Authenticated Device. The authenticated device passes this test if this data is accessible for no less than a total of 30 consecutive days from the start of the test.
Independently designed transmitter constraints	Success of *Independently designed transmitter constraints* is acheived if and only if the transmitter is commercially designed or the transmitter/receiver is built independently such that the Part 2 of FCC 47 CFR section 2.1033, 2.1046, and 2.1057 constraints are satisfied, and the transmitter/receiver network adheres to the FFC 47 CFR 2.911 authorization constraints.
Integrated commercial transmitter/modem constraints:	Success of *Integrated commercial transmitter/modem constraints:* is acheived if and only if the transmitter is independently designed or the device is commercially designed such that the datasheet of the commercial model confirms that the transmitter is authorized according to the *FCC KDB 996369 D01: Transmitter Module Equipment Authorization Guide*

Measures of Success for Microcontroller

Specification/Constraint	Success Condition
TCS-MS1: The customer, Dr. Charles Van Neste, specifies that the Telluric Current Sensor may use the Particle Boron Microcontroller as the combined MCU and Telecommunication modules.	This specification was nullified due to Particle Boron being end of phase
TCS-MS2: The customer, Dr. Charles Van Neste, specifies that the MCU must use a Fast Fourier Transform algorithm to extract the frequency amplitudes of the first twenty 60 Hz harmonics of the measured telluric currents	Success of **TCS-MS2** is achieved if and only if the microcontroller passes its three tests ten times. Test (1) requires that a predetermined signal containing a known, precise, and randomized Fourier spectra with a bandwidth of at minimum 10 MHz will be sampled at 50,000 samples per second. Test (2) is that sampled data will be fed into the MCU to be processed by the FFT algorithm. If the FFT outputs the 60 Hz, 120 Hz, ..., 1200 Hz amplitudes, and the sum of the squared errors between the theoretical and calulated FFT amplitudes does not exceed 0.01% of the average theoretical amplitude of the spectra over the 1200 Hz, the FFT passes the accuracy test. Test (3) is that if the FFT processes the amplitudes in 30 seconds, it passes the speed test.
TCS-MS3: Dr. Charles Van Neste specifies that the MCU shall send the following data per the following procedures to the Telecommunication Subsystem...	Success of **TCS-MS3** is achieved if and only if the microcontroller passes the tests for **TCS-MS3.1 TCS-MS3.2**
TCS-MS3.1: During regular operation (i.e. no fault detected), the MCU shall send a transmission packet every hour containing the 20 FFT extracted amplitudes calculated stored separately in five minute increments from 60 minutes prior to transmission until 5 minutes prior to transmission. The MCU shall also send a boolean indicator value of FALSE indicating the lack of a fault.	Success of the **TCS-MS3.1** specification will be acheived if and only if during a 24-hour test, the MCU shall successfully send 600 FFT amplitudes (floats) and 1 boolean indicator of false as a transmission packed every hour. Under idealized transmission conditions, success will be achieved if and only if (1) the data is received with 99% accuracy and (2) it is received in less than 5 minutes after process starts

TCS-MS3.2 Within the first 10 minutes of detection of an abnormality in the measurement data*, the MCU shall send a priority transmission packet to the telecommunications unit containing the 20 FFT extracted amplitudes stored separately for - at the largest division of time - every one hundredth of a second between the time of detection and 60 seconds after detection. The MCU shall also send a boolean indicator value of TRUE indicating the presence of a fault.	Success of the **TCS-MS3.2** specification will be acheived if and only if during a 24-hour test, under idealized transmission conditions, success is achieved if and only if (1) six hypothetical faults are sent, each containing a transient value (100 x the data array) that flags the indicator,(2) upon detection, a priority transmission is received containing 10,000 FFt amplitude graphs and a boolean value of TRUE with 99% accuracy, and (3) the transmission is received in 10 minutes
NERC Section B 1.2.5	Success of **NERC Section B 1.2.5** is acheived if and only if device will boot arbitrary, modified, or malicious code if an attacker manages to flash it over-the-air (OTA) or via a physical interface. This can be done by taking, a known working compiled binary, modifying a single byte in the code, attemping to apply this binary code to the MCU and testing to see if it flags the update or not.

Power Measure of Success:

Specification/Constraint	Success Condition
TCS-PS1 The customer, Dr. Charles Van Neste, specifies that the system must be powered locally and require power system maintenance to be done as infrequently as possible (4 battery changes a year at maximum).	Success of **TCS-PS1** will be conducted by an experiment with the chosen source of power by connecting the expected load outcome over a 2-week period of time. Then calculate the daily discharge rate for the 14-day period, once found scale it by the 90 day specifictaion asked by the customer.
TCS-PS2 The customer, Dr. Charles Van Neste, specifies that the system must incorporate surge protection devices complying with IEEE Std C62.41 to withstand high-voltage transient overvoltage's.	Success of **TCS-PS2** will be conducted by an experiment using the chosen surge protection equipment for the TG-FCI. Hook the chosen equipment to a controlled surge generator and measure the secondary side of the power system with an Oscilloscope to prove voltage spikes were stopped from passing through. (Secondary side will show 0.0V on Oscilloscope to prove voltage spikes were negligible on the secondary side.)
TCS-PS4 The customer, Dr. Charles Van Neste, specifies that the output of the regulators and LDOs meet low-noise requirements, ensuring no measurable power supply leaks into the sensor's bandwidth.	Success of **TCS-PS4** will be conducted by disconnecting all power sources and external connections, set a digital multi-meter (DMM) to its highest resistance range. Place DMM probe on the primary ground and the other probe on the secondary (isolated side) ground. The multimeter must read infinite resistance, confirming that no hardware compromises the high-voltage isolation rating.

Power Subsystem Measures of Success

- **Autonomous Uptime:** Demonstrate at least 24 hours of continuous operation under full active load as a baseline, and support unattended field deployment for at least 3 months (90 days) [SR-04].
- **Galvanic Isolation and Transient Protection:** Demonstrate a verified 2kV RMS isolation barrier between external power inputs and internal instrument grounds to effectively eliminate ground loops and protect downstream circuitry.
- **Power Quality and Noise Suppression:** Verify that the output of the regulators and LDOs meet low-noise requirements, ensuring no measurable power supply leaks into the sensor's bandwidth [SS-7],[SS-22].

Weatherization Measures of Success

- **Temperature Resistance:** Success shall be achieved if the weatherproofing system maintains its structural integrity and electrical functionality throughout an operating temperature range of −30°C to 60°C.
- **Waterproofing Verification**- Success shall be accomplished if moisture-indicating test strips placed inside the enclosure show no evidence of water intrusion following a 60- minute water exposure test.

3.

· Critique:

Requirement	5: Strongly Agree (Exceeds)	4: Agree (Achieves)	3: Neutral (Generally Meets)	2: Disagree (Below)	1: Strongly Disagree (Significantly Below)	Rating
P5. Constraints Listed constraints from governing bodies, standards organizations, or the community?	Applicable standards and regulations cited by number, with how each constrains the design. Sponsor and site constraints included.	**Applicable standards cited by number. Effect on design stated briefly.**	**Standards bodies named (IEEE, UL, OSHA) without specific documents.**	Constraints limited to budget and deadline.	No constraints listed.	**3.5**

While no changes were necessarily made to project proposal, other than a few deletions of redundant/unapplied standards, constraints were evaluated and applied in greater detail in the conceptual design Ethical Considerations subsection.

Conceptual Design Revisions:

1.

- Critique:

Part B: Conceptual Design						
C1. Principles, Models, Tools Described the fundamental principles, mathematical models, and tools used?	Governing principles and equations stated. First-order model with calculated values. Each tool named with what it will verify.	Principles and tools named. Models partly unquantified.	**Principles named in general terms. No model or calculation.**	Course names or software listed only ("circuits knowledge and MATLAB").	Not addressed.	**3**
Instructor: Relevant equations and tools are presented, but most equations are not evaluated with design values and some claims are unsupported. **GTA:** No models or calculation for DAQ						

- Text Before Revision:

3. Battery State of Charge and Autonomous Runtime

$$T_{\text{runtime}} = \frac{C_{\text{batt}} \cdot \text{DoD} \cdot \eta_{\text{sys}}}{I_{\text{avg}}}$$

Where:

- C_{batt} is battery capacity (Ah).
- DoD is maximum depth of discharge (0.80 for LiFePO$_4$).
- η_{sys} is power conversion efficiency (0.85).
- I_{avg} is average continuous current (A).

5. LDO Ripple Attenuation and PSRR

An ultra-low-noise LDO, such as the LT3042, downstream of the isolated switching converter attenuates switching-frequency ripple before it reaches the analog front-end:

$$V_{\text{ripple,out}} = V_{\text{ripple,in}} \times 10^{-\text{PSRR (dB)} / 20}$$

Where:

- $V_{\text{ripple,in}}$ is peak-to-peak ripple voltage entering the LDO from the switching converter or isolated module (mV).
- PSRR is the LDO's power supply rejection ratio at the switching frequency (e.g., at least 75 dB at 100 kHz).
- $V_{\text{ripple,out}}$ is the resulting ripple on the analog power rail (e.g., μV).

-DAQ was not included prior to revisions.

- Text Post Revision:

3. Battery State of Charge and Autonomous Runtime

$$T_{\text{runtime}} = \frac{C_{\text{batt}} \cdot \text{DoD} \cdot \eta_{\text{sys}}}{I_{\text{avg}}}$$

Where:

- C_{batt} is battery capacity (20.1 Ah for Voltaic V75 Always On pack).
- DoD is maximum depth of discharge (0.85).
- η_{sys} is power conversion efficiency (0.85).
- I_{avg} is average continuous current (A).

Design Evaluation:

Using the Voltaic V75 Always On battery pack ($20,100\ \text{mAh}$ / $72.36\ \text{Wh}$) to eliminate low-current auto-shutoff during MCU sleep cycles:

1. **Continuous Active Load Profile (24-Hour Baseline Target):** Under full continuous active load and active cellular transmission ($I_{\text{avg}} = 0.575\ \text{A}$):

$$T_{\text{runtime, continuous}} = \frac{20.1\ \text{Ah} \times 0.85 \times 0.85}{0.575\ \text{A}} = 25.25\ \text{hours}$$

   (Meets the 24 hour run time requirement).

2. **Field Operations (3-Month / 90-Day Operational Target):** Under a 1% duty-cycled telemetry regime (about 36 s of active transmission per hour at $0.575\ \text{A}$, with deep-sleep background monitoring of $0.93\ \text{mA}$ for the remaining time), the average current draw is $I_{\text{avg, duty}} = (0.01)(575\ \text{mA}) + (0.99)(0.93\ \text{mA}) \approx 6.67\ \text{mA}$:

$$T_{\text{runtime, duty}} = \frac{20.1\ \text{Ah} \times 0.85 \times 0.85}{0.00667\ \text{A}} = 2177.3\ \text{hours}\ (\approx 90.7\ \text{days} \,/\, 3.02\ \text{months})$$

   (Fulfills the 3-month operations).

5. LDO Ripple Attenuation and PSRR

An ultra-low-noise LDO, such as the LT3042, downstream of the isolated switching converter controls switching-frequency ripple before it reaches the analog front-end:

$$V_{\text{ripple,out}} = V_{\text{ripple,in}} \times 10^{-\text{PSRR (dB)}/20}$$

Where:

- $V_{\text{ripple,in}}$ is peak-to-peak ripple voltage entering the LDO from the switching converter or isolated module (mV).
- PSRR is the LDO's power supply rejection ratio at the switching frequency (e.g., at least 75 dB at 100 kHz).
- $V_{\text{ripple,out}}$ is the resulting ripple on the analog power rail (μV).

Design Evaluation: With an incoming switching ripple of $V_{\text{ripple,in}} = 50\ \text{mV}_{\text{p-p}}$ and an LT3042 PSRR of $75\ \text{dB}$ at $100\ \text{kHz}$:

$$V_{\text{ripple,out}} = 50\ \text{mV} \times 10^{-\frac{75}{20}} = 50\ \text{mV} \times 0.0001778 = 8.89\ \mu\text{V}_{\text{p-p}}$$

(Meets the requirement for $< 10\ \mu\text{V}_{\text{RMS}}$ analog noise floor).

7. LDO Power Dissipation and Thermal Limit

Because regulators dissipate excess voltage as heat, calculate the LDO's power dissipation:

$$P_D = (V_{\text{in}} - V_{\text{out}}) \cdot I_{\text{load}}$$

The resulting semiconductor junction temperature is:

$$T_J = T_A + (P_D \cdot \theta_{JA})$$

Where:

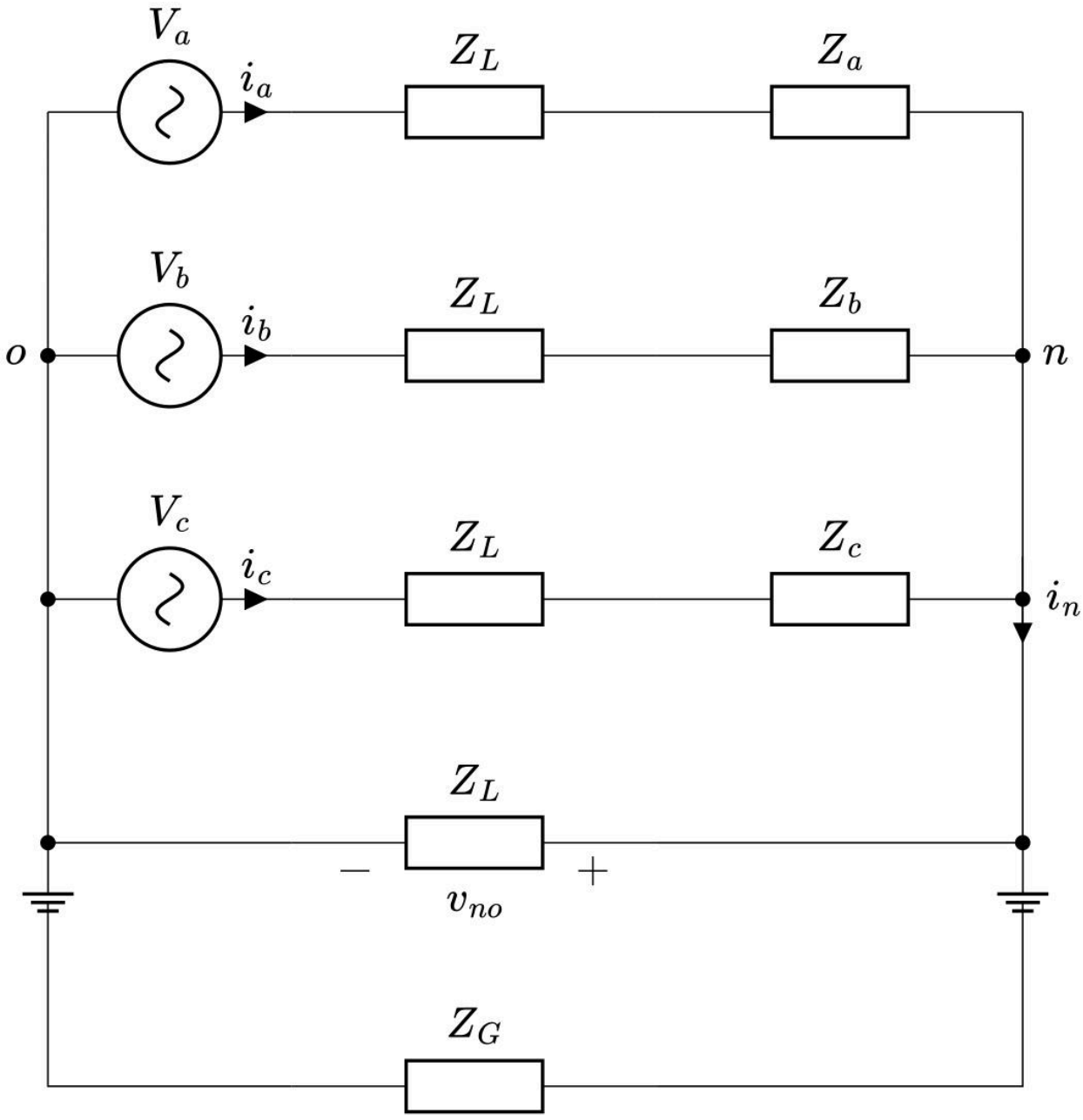
- P_D is power dissipation (W).
- V_{in} and V_{out} are the LDO input and output voltages ($5.0\ \text{V}$ and $3.3\ \text{V}$).
- I_{load} is continuous current drawn by the analog section ($0.100\ \text{A}$).
- T_A is ambient temperature inside the enclosure (50°C).
- θ_{JA} is the IC package's junction-to-ambient thermal resistance (41°C/W for MSOP-10).

Design Evaluation:

$$P_D = (5.0\ \text{V} - 3.3\ \text{V}) \times 0.100\ \text{A} = 0.17\ \text{W}$$

$$T_J = 50^\circ\text{C} + (0.17\ \text{W} \times 41^\circ\text{C/W}) = 56.97^\circ\text{C}$$

(Safely below the LT3042 maximum operating junction limit of 125°C*).*


V_a
i_a
Z_L
Z_a
V_b
i_b
Z_L
Z_b
o
n
V_c
i_c
Z_L
Z_c
i_n
Z_L
$-$
$+$
v_{no}
Z_G

To prove this, the following derivation is provided. First consider that the path through the ground may be represented as a wire with impedance Z_G. Apply Kirchhoff's current law at the node n:

$$i_a + i_b + i_c + i_n = 0$$

Solve for the neutral-neutral current for an unbalanced system:

$$i_n = -\left(\frac{v_a}{Z_a} + \frac{v_b}{Z_b} + \frac{v_c}{Z_c}\right)$$

Apply current divider to find current through the soil:

$$i_G = i_n \frac{Z_L}{Z_L + Z_G}$$

Recall the voltage distribution in a medium:

$$V(r) = \frac{\rho I}{2\pi r}$$

In this instance, the current in the ground is what produces the voltage distribution, so:

$$V_G(r) = \frac{\rho I_G}{2\pi r}$$

In measuring a step voltage, r is displaced by a distance s and the voltages are subtracted:

$$V_{step}(r) = \frac{\rho I_G}{2\pi}\left(\frac{1}{r} - \frac{1}{r+s}\right)$$

Thus, given the model, the step potential will decrease non-linearly with radius but linearly with I_G meaning the data will be a linear scale of the current.

2.

- Critique:

C2. Alternatives Proposed alternative solutions with pros and cons?	Three or more distinct alternatives in a weighted decision matrix tied to the specifications. Selection justified.	Two or three alternatives with pros and cons. Selection justified qualitatively.	Two alternatives. Pros and cons not tied to specifications.	**One real option plus a strawman, or no pros and cons.**	Single solution only.	**2**
GTA: No comparison for power, inconsistencies for communication, missing for DAQ						

The DAQ subsystem was fully written with extensive comparative analyses (see below.) All decision matrices were rewritten for all subsections to account for the new quantitative specifications and constraints (*See new commit changes.)

- Text Before Revision:

N/A

- Text Post Revision:

Data Acquisition Unit (DAQ)

This section provides information pertaining to the identification and justification of choices made in designing the DAQ. In general the DAQ includes measurement of data, signal conditioning, and digital conversion. Many of these do not have multiple valid solutions. For example, in the signal conditioning phase, there will be an unbalanced PI attenuator which matches the input and output impedance while attenuating the signal. Such a small and rather trivial step of signal conditioning will not need a decision matrix. As a result, the copraisons will be: the different materials and arrangement of the electrodes, the different possible reference points, and the analog to digital converter.

According to the specifications (inluding power and monetary budget), the DAQ must have:

- measurements taken by the voltage difference using two or more electrodes **TCS-DS1**
- accuracy within 10% (**TCS-DS1.1**) or 20% (**TCS-DS1.2**) of the true RMS voltage.
- compatibility with the MCU in all events (excluding lightning strikes or UHV transients) **TCS-DS2**, **TCS-PC2**
- a sampling rate greater than 10 kHz **TCS-MS2.1**
- a power consumption less than 20W according to the power budgetting **TCS-PS1**, **TCS-F2.1**
- a price less than (or optimized under) $100 according to the budget **TCS-F1**

see budget for power and money budgetting specifics past what is defined by the customer

Each decision made for the DAQ will be supported by these specifications (no constraints directly apply to this but moreso to the other aspects, especially the power supply an distribution).

Comparison of Electrode Materials and Shape

The first decision to make is the material of the electrodes. The electrode must help the DAQ have:

- measurements taken by the voltage difference using two or more electrodes **TCS-DS1**
- accuracy within 10% (**TCS-DS1.1**) or 20% (**TCS-DS1.2**) of the true RMS voltage.
- compatibility with the MCU in all events (excluding lightning strikes or UHV transients) **TCS-DS2**, **TCS-PC2**
- a price less than (or optimized under) $100 according to the budget **TCS-F1**

The accuracy of the electrodes will be due to the patina and resistivity of the material. Resistivity will be ranked low because it will be only a scalar change whereas patina is a little more complicated. The compatibility in all conditions (which incluse moist or dry soil) will be largely determined by the patina. As a result of these two factors, patina will be ranked the highest. The cost of the electrodes may be a large portion of the DAQ budget, so it will be weighted decently high but still not higher than the patina.

Metric	Weight
Patina	50%
Cost	40%
Resistivity	10%

Now, the material of the electrodes will be considered. Of course, it must be conductive and withstand the environment well (because they cannot be shielded so that they may conduct). Corrosion resistant metals include: aluminum, copper, brass, bronze, stainless steel, galvanized steel and titanium. Each of these develops a patina or surface oxide which changes the conductivity but some corrode faster or to a greater detriment in soil. References: Patina [AB-1], [AB-2], [AB-3], [AB-4], [AB-5]; Cost [AB-6]; Resistivity [AB-7].

Material	Patina (0.5)	Cost (0.4)	Resistivity (0.1)	Total
Aluminum	1	4	4	**2.5**
Copper	2	3	5	**2.7**
Brass	2	3	4	**2.6**
Bronze	2	2	4	**2.2**
Stainless Steel	3	5	3	**3.8**
Galvanized Steel	1	5	3	**2.8**
Titanium	0	0	3	**0.3**

Now it is clear that stainless steel is the best option of the materials considered. Next, the decision must be made on how the data will be collected. Any method must measure the potential difference through the soil to find the electric field. Because all of the ways to extract voltage from the earth rely on implanted conductors, the only decision to be reached is the shape and the material. The shape, or burial configuration, is what determines the measurable data points and the options considered were: delta, wye, collinear rods, and parallel plates. The goals of the arrangement of the electodes is to minimize the complexity and cost while maximizing the data. This implies that the metrics must be the complexity of integration (which includes the burying process), data quality, and cost. Data quality is the sublime metric for the shape because poor quality data is useless data, so it is weighted double the others.

The delta configuration is formed by three electrodes in an equilateral triangle; this would give us three voltage differences that form a loop (helpful for Kirchoff's Equations). Overall, the difficulty of integration is determined by wiring three electrodes and burying three electrodes (3). The data quality would be fairly good, although it gives only two linearly independent voltages ($V_{AB} - V_{BC} = -V_{CA}$) (3). The cost is better than the wye variant but not as good as the collinear rods (4).

The wye configuration is the same as the delta, but there is an electrode place in the center of the delta; this would give us three linearly independent voltages that form many closed loops. The complexity of implementation is slightly higher than the delta configuration (3). The data quality is extremely good compared to the other candidates because of the number of independed equations (5). The cost is again slightly more than the delta configuration with the addition of another electrode (3).

The collinear rods would give one voltage reading. This is very simple and cheap but not entirely cost effective. The complexity is spectacular and as simple as it could be (5). THe data quality is poorer than the delta configration (2). THe cost is as good as it could be (5).

The parallel plates would gives us only one voltage reading as well. The complexity is still very high, and the cost as well (2 and 2 respectively). There was uncertainty whether or not the voltage difference may self-induce an electric field across the plates, but it wasn't worth testing as it results in the same data points as the rods at best (1).

Shape	Complexity (0.25)	Data Quality (0.5)	Cost (0.25)	Total
Delta (Δ)	4	3	4	**3.5**
Wye (Y)	3	5	3	**3.75**
Collinear Rods (· ·)	5	2	5	**3.5**
Parallel Plates (‖)	2	1	2	**1.5**

Overall, the stainless steel wye configuration was shown to be superior to the other options. Stainless steel is about $2/kg so the steel rods as a whole will cost $10.

Comparison of Electrode Reference Point

By Kirchhoff's voltage law, the voltage between A and B is the same voltage as the voltage between A and B through any point. This means that the line voltage readings are entirely independent of whatever reference point is taken to read the phase voltages. Because of this independence, it is preferable not to directly measure the line voltages because this reduces the number of data points, though this option ought to be considered for a simplistic possibility. This reference point could be N the central node, it could be the grounding rod off of the neutral line, n, or it could be some point R many meters away that represents close to 0 voltage with respect to the rest of the earth (called remote-earth). All three possibilities will be considered and weighted in distinct categories.

These categories must depend firstly on the specifications: the electrode reference point must help the DAQ have:

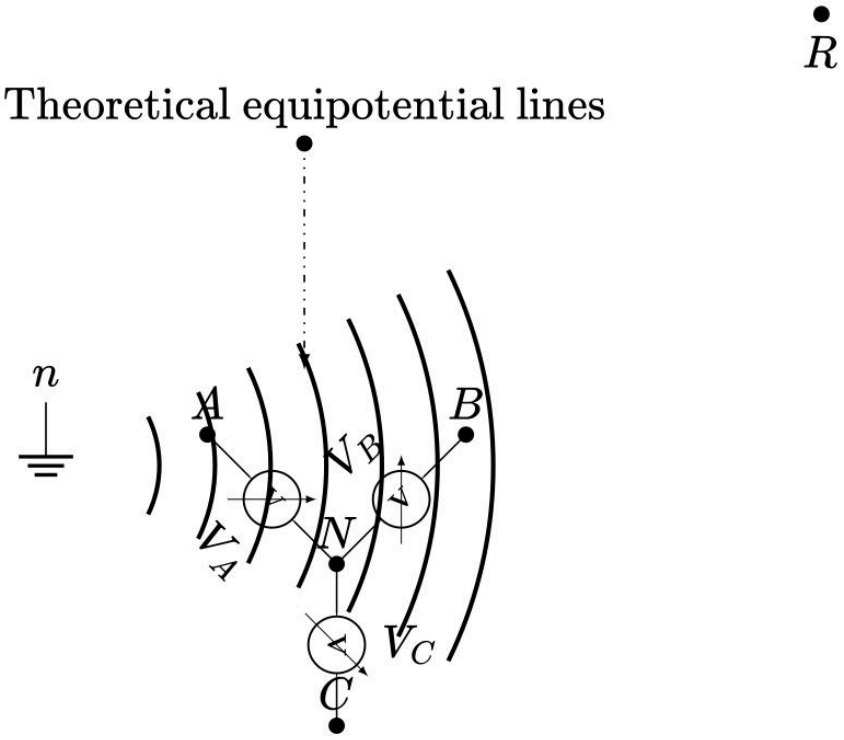
- measurements taken by the voltage difference using two or more electrodes **TCS-DS1**
- accuracy within 10% (**TCS-DS1.1**) or 20% (**TCS-DS1.2**) of the true RMS voltage.
- a price less than (or optimized under) $100 according to the budget **TCS-F1**

In addition to these specifications, the electroe reference point will provide insight and clarity the what can be analysed, so this will be considered as well. Accuracy entails the elimination of error from the readings whereas precision is the elimination of noise. All reference points are equally accurate, not inhenertly the same noisiness (this will be discussed), depending on the processing later so this category will be of no weight and uncounted since it will not affect the scores. Noise is weighted low because of the general equality that each reference node provides, so the score can be a little more sensitive. The next goal is to produce data which is insightful (hence the reason why the reference ground was abandoned), this will be elaborated on in each respective section because each produce their own insight. This weight for insight will be fairly high considering the impact it could have. The reference points must also be attainable in application. From this they ought to be cost effective, simple to install, non-invasive, and maintainable. Cost effectiveness will be rated at a low medium this is because the solution still ought to be cheap, but the primary cost will come from the electronics, not necessarily the electrode scheme. Ease of install will be rated low not because it's unimportant, but because they are relatively similar in ease. Non-invasive will be weighted medium for similar reasons; all three options are not invasive so they're fairly equal, but node n will propose some stray from this. Maintainability will be ranked medium-high because a large importance of this project is the longevity of it.

The selected weights and categories are provided each will be ranked 1-5 and summed:

Metric	Weight
Accuracy	0%
Noise	10%
Insight	35%
Cost	12%
Installation	8%
Non-invasive	15%
Maintainability	20%

Now that there is a metric by which to determine the reference node, an in-depth analysis of such will follow. In parenthesis, the number will represent the score in the discussed category.



Reference N

This reference is the center of the inlaid electrodes A, B, and C. This forms the neutral equivalent of the pseudo-wye model. The resulting data would be V_{AN}, V_{BN}, and V_{CN}, three vectors which differ greatly in angle (120° apart). The obvious benefits of this measurement scheme is that all the points are condensed; the housing of the electronics could sit just above the neutral embed and reduce the square footage of the device greatly. The model is also easily interpretable thanks in part to the similarities to the standard wye formation. This also allows the opportunity for a voltage (V_{AN} for example) to be parallel to the radius of the current wave propagation from n, forcing another (V_{BC}) to be perpendicular to it.

This would give insight into the general shape of the voltage (and therefore current density) spatial distribution. This insight is particularly beneficial for goals that lie at the edge of the scope of this project: analyzing where the fault happened based on spatial current density distribution (5). Maintenance would be very similar to maintenance of the other three electrodes given that it occupies the same space (4). This method is minimally invasive due to the separation from the power line and connecting wires (5). The cost is the same as adding another electrode and is the cheapest of the reference point options again because it is so close to the other electrodes (4). The noise of this method may be higher than the others. Realistically, there is no certainty for lack of data in measuring grid induced telluric currents, but it is clear that since this node will cover less distance, it will have a lower voltage reading. The lower the voltage reading, the closer it could be to the noise RMS (3). Installation would be relatively simple, as simple as the other three electrodes (4).

Reference n

This reference is the grounding electrode of the three phase power lines. The resulting data would be V_{An}, V_{Bn}, and V_{Cn}, three vectors which differ slightly in angle and magnitude. The benefit of this measurement scheme is the relation to the neutral line. This could give valuable data on the balance of the three phase power system. Effectively, it sets the neutral line as the center of a voltage distribution map of which three points are selected. This would also help with spatial modeling of the current density in a uniquely different way than node N.

This insight would be very valuable (5) (notice that even though the reference points are uniquely different, they're both equally insightful, this will be considered later). Maintenance would be slightly more complicated but already considered; the agency which maintains power lines already maintains the neutral line's grounding node (4). This method is slightly more invasive than any other although it is still very much outside of the transformer and not on the power pole so it remains very non-invasive (4). The cost is minimal as the neutral electrode is already there (5). The noise of this system could be very low if the electrodes are buried far away, but the farther away that they are, the closer their data points become making it obsolete to have many. As a result, this system would have to be closer (and therefore noisier) to read unique data across all three electrodes. In likelihood, the noise would be similar the previous option (3). Installation is not a concern because it will already be there (unless the device is in an area where there is no ground connection). This final note is of great concern to \textbf{Seal Team 2} because it restricts where this device can be placed, as a result it will score a 1 in this category so as not to fully condemn the idea but very much discourage it (1). It does leave open an option which will be discussed in the decision section however.

Reference R

This reference is some buried electrode a substantial distance away from the measurement site. The data obtained would be V_{AR}, V_{BR}, and V_{CR}. The benefit of this reference node is that since, in theory since there are no single point voltages, $V_{AR} = V_A - V_R$ and $V_R \approx 0$ because of its distance from any voltage distribution. This means the representation obtained is close to the absolute voltage at that point. This drastically reduces the noise introduced into the system and makes calculation for spatial distribution much easier.

The insight provided would be very valuable to voltage distribution mapping and power line health (5). The maintenance would be more difficult because every meter away that this electrode is, the more area this device takes up (3). This method is minimally invasive (5). The likelihood of the wire needed to be buried is decently high which would greatly increase the cost (3). There would be as low noise as possible with this reference scheme (5). The installation as mentioned may be difficult and costly depending on the accuracy desired (3).

Decision

Below is the table showing the results outlining the decision of which reference node to use.

Reference Node	Insight	Maintenance	Non-invasive	Cost	Noise	Installation	Total
N	5	4	5	4	3	4	**4.4**
n	5	4	4	5	3	1	**4.13**
R	5	3	5	3	5	3	**4.2**

As previously alluded to, there is something considerable about these reference ideas: none of them are exclusive. This means it is feasible, though not cost effective, to implement multiple or all of these references. For now, the product will be designed around node N, but an option may come available for multiple references or selecting based on application. Such decisions could be: node N within a city for small square footage, node n in rural areas where spatial distribution would be massively helpful, node R near substations where the noise threshold is very high. Regardless, node N has been selected as the best candidate for general prototyping.

Comparison of Analog to Digital Converters (ADCs)

There will be additional circuitry used to condition the signal before it reaches the ADC, but much of that is trivial and straightforward, so much so that there is only one effective way of doing those things. As a result they will be discussed later. The next analysis is to compare different ADCs.

The ADC must help the DAQ have:

- measurements taken by the voltage difference using two or more electrodes **TCS-DS1**
- accuracy within 10% (**TCS-DS1.1**) or 20% (**TCS-DS1.2**) of the true RMS voltage.
- compatibility with the MCU in all events (excluding lightning strikes or UHV transients) **TCS-DS2**, **TCS-PC2**
- a sampling rate greater than 10 kHz **TCS-MS2.1**
- a power consumption less than 20W according to the power budgetting **TCS-PS1**, **TCS-F2.1**
- a price less than (or optimized under) $100 according to the budget **TCS-F1**

Because of specification **TCS-DS2**, additional measures are taken to provide an alternate path for high voltage spikes in the event of a ground fault this will be refered to as the surge path. Because the surge path will have a greater signal to noise ratio naturally because the voltage will increase but the noise threshold will not, the ADC for the surge path can sacrifice a *little* noise for a higher sampling rate and badwidth while preserving power and monetary cost. IN general, **TCS-DS2** and **TCS-PC2** are accomplished through the division of paths into main and surge path and in the protection circuitry in the event of a lightning strike.

Now from the goals the parameters can be found all of which (unless otherwise stated) will be ranked 1-5 insufficient to profiecient. By **TCS-DS1**, the ADC ought to have high resolution and low noise. The resulting parameters will be resolution, RMS noise, effective number of bits (ENoB), and common-mode rejection ratio (CMRR). These parameters dictate the ADC's ability to remain accurate and precise given a range of values. By **TCS-DS2**, the ADC must be compatible with the MCU so each selected ADC must have SPI capabilities, this will be an unweighted binary metric as any ADC which violated this will violate the specification and thus not be considered. By **TCS-MS2.1**, the sampling rate must be greater than 6 kHz. For accuracy and the opportunity of oversampling and averaging, this will not be a binary metric. By **TCS-PS1**, **TCS-F2.1**, and the *budget* section, the ADCs combines must consume less than 17 W, but any minimizing of this while preserving the other metrics is desirable. By **TCS-F1** the price must be less than $60 for all ADCs combined, this may also be minimized if possible while preserving other metrics. By metric **TCS-DS2**, the bandwidth must be great to include the harmonics of a surge or high voltage transient, applicable only for the surge path. In general, there will be three measurements taken at all times which corresponds to six ADC channels (main and surge ADCs for each signal), so the cost per channel will be considered.

Special attention will be focused on specifications which will be collected for each ADC but will not be factored in to the decision process. This, in general, is because they're all equal (in the case of the binary SPI compatability metric) or because they don't have any effective difference on the system. Such metrics of the latter are the conversion rate (not linearly independent from sample rate), common-mode range, and differential range (both because the signal conditioning will put the sigal within the range that the ADC can handle).

The metrics are: Resolution, Sample Rate, Cost per channel, Power, RMS Noise, ENoB, CMRR, and bandwidth (only for surge ADC). Now the weight need to be established for each of the metrics. The weights and ranking will change between the main and surge ADCs. For example, the sample rate of 250 kHz would score a 5 for the main branch because that provides 25 averaged samples at 10 kHz if oversmapling is implemented. On the other hand 250 kHz may not capture many harmonics of a fast voltage transient so it would score a 3 for the surge branch.

Resolution is an important metric because it determines how accurately data is gathered, but because the increase of bits results in an exponential growth of the resolution, most ADC choices will revolve around the 10-16 bit range. Still, resolution is very important and will be weighted highly for both paths. Sample rate is important so long as it is above 10 kHz but after 50 kHz all options are fairly similar. Moderate weight will be put to this metric, but the ranking will be very extreme (0 for <6 kHz 1 for <10 kHz and 5 for >50 kHz). Cost per channel is fairly important because with 3 channels per branch, the average price per channel must be <$10. This will be weighted high. Power is also decently important because of the specification applied, but this restriction is not as tight as the cost. This will be weighted medium. Noise as a whole will take a high weigh as well but each individual secion will be lower weighted so their sum is high but not overpowering. RMS Noise, the measurement of how much voltage must be applied to be distinguished from noise, will be the heghest weight of the noise categories because of the precision needed for 1% error. ENoB will be of moderate weight. Bandwidth will be of high weight for the surge ADC. CMRR will be of high weight as well becuase the common-mode drift must be taken into account by our measurement scheme (central node wye).

Main ADC Metrics	Weights
Resoution	20%
Sample Rate	15%
Cost per Channel	15%
Power	15%
RMS Noise	20%
ENoB	5%
CMRR	10%

Overall, the precision of the device is weighted slightly lower because of the voltage amount which will be measured. The voltage will be sufficiently far from the noise threshold and therefore require less precision to separate it from noise. ENoB is not considered fro these wieght because of the minimal affect it would have on the values (<1%)

Surge ADC Metrics	Weights
Resoution	20%
Sample Rate	15%
Cost per Channel	15%
Power	15%
RMS Noise	10%
CMRR	5%
Bandwidth	20%

Decision

After searching, **Seal Team 2** selected 6 relevant ADCs [datasheet reference]:

- AD7352 [AB-8]
- LTC1408-12 [AB-9]
- MCP3460R [AB-10]
- ADS130B04-Q1 [AB-11]
- MAX1088 DS [AB-12]
- MCP33151X-14X-025 [AB-13]

Below is the condensed table showing all of the relevant values to the decision matrices recall that the omitted metrics are the conversion rate, SPI compatibility, common mode range annd differential input range.

ADC	AD7352	LTC1408-12	MCP3460R	ADS130B04-Q1	MAX1088 DS	MCP33151X-14X-025
Resolution (bits)	12	12	16	16	10	14
Sample Rate (kS/s)	3000	100	153.6	32	150	250
Channels (#)	2	6	1	4	1	1
Cost per channel ($)	9.31	3.83	1.35	2.365	12.92	1.3
Power (mW)	26	15	2.2-6.8	6	0.5-0.75	1
Input Referred RMS Noise (μV RMS)	385	122	36.6	36.62	1290	231
Effective Number of Bits (#)	11.5	11.67	16	16	9.84	13
Bandwidth (kHz)	77000	5000	150	70	100	200
Common-Mode Rejection Ratio (dB)	100	83	122	128	100	84

Below are the decision matrices depicting first the ADC optimized for the main path and the surge path respectively.

ADC (*Main Path*)	AD7352	LTC1408-12	MCP3460R	ADS130B04-Q1	MAX1088 DS	MCP33151X-14X-025
Resolution	4	4	5	5	3	5
Sample Rate	5	4	5	3	5	5
Cost per Channel	3	4	5	4	2	5
Power	2	2	3	3	4	3
RMS Noise	3	4	5	5	2	4
ENoB	4	4	5	5	4	5
CMRR	4	3	5	5	4	3
Total	**3.5**	**3.6**	**4.7**	**4.25**	**3.25**	**4.3**

The selected ADC for the main path is the MCP3460R which is $1.35 for one channel and 6.8mW max per channel. The largest impact this could make in the budget is $4.05 and 20.4mW.

ADC (*Surge Path*)	AD7352	LTC1408-12	MCP3460R	ADS130B04-Q1	MAX1088 DS	MCP33151X-14X-025
Resolution	5	5	5	5	4	5
Sample Rate	5	3	3	0	3	3
Cost per Channel	3	4	5	4	2	5
Power	2	2	3	3	4	3
RMS Noise	5	5	5	5	4	5
CMRR	4	3	5	5	4	3
Bandwidth	5	5	3	1	3	3
Total	**4.2**	**4**	**4**	**0**	**3.35**	**3.9**

As the reader can see, then inability of the ADS130B04-Q1 to sample high enough to capture a fast transient has disqualified it with a 0. The selected ADC for the surge path will be the AD7352 which is 9.31 per channel (but it comes as a pair of channels so 2 ADCs would be bought resuting in $18.62) and 26mW per ADC. The maximum impact this could have is $18.62 and 52mW.

3.

- Critique:

C3. High-Level Details Explained the high-level details of at least one solution?	End-to-end operation, role of each subsystem, interfaces (voltage levels, signals, protocols), and subsystem owners explained.	Clear explanation. Some interfaces undefined.	**Explanation present but some subsystems are black boxes.**	**Restates the problem. Little on how the solution works.**	Not addressed.	**2.5**
Instructor: Selected architecture is described, but the DAQ/analog front end is substantially underdefined and several interfaces remain unresolved. **GTA:** Missing for DAQ						

- Text Before Revision:

N/A for DAQ

- Text Post Revision:

Atomic Subsystem Specifications

Data Acquisition

Now begins the in depth specifications of the DAQ. First will be a general flow of the main stages, then a rigorious exaplanation of ease stage will follow.

In the DAQ, there are four main stages: raw data retrieval, protection (which will involve pathing explained later), signal conditioning, and analog to digital conversion. Raw data retrieval is simply acquiring the voltage in the ground. The protection will include limiting the current, a fuse for UHV surges, and a way to separate high and low voltage paths. The signal conditioning includes the attenuation of the signal so that it can be filtered for the desired range then amplified again to match the range of the ADC. The analog to digital conversion is just the ADC sampling.

Raw Data Retrieval

Of all the phases, this one is the most straightforward. The requirement is that the DAQ shall read the grid induced telluric voltages using at least two electrodes. It is necessary that these elctrodes be separated by some distance that provides enough amplitude (voltage drop is proportional to radius, 1m suggested). By the decision made earlier, these electrodes shall be made of stainless steel for its resistivity, cost, and low impedance patina. By another decision made previously, these electrodes shall be configured in a wye formation with three of the electrodes be configured as the three points of an equilateral triangle and the fourth be the very center of such triangle. The voltages measured shall be three: each point of the triangle to the center.

The imbalance of the three phase power in the power lines will produce a current in the soil which implies the presence of a voltage gradient in the soil. The electrodes shall input this voltage (or some scalar shift/scale of it) into the next phase.

DAQ Protection and Pathing

The DAQ shall be protected from surges over 500 V and overcurrents which is the primary goal of the protection phase. Each node shall have a resistor on it which sufficiently prevents overcurrents (10k). The first path shall clamp at (or close to) 10 V so that the path is saturaded, not opened. Same will be for any voltage above 500 V and the second path. These clamps shall be capable of conducting within $30\mu s$, the standard period of a lightning strike surge.

The rest of the circuit shall also be protected if there is leftover voltage after the initial protection stage. Before the amplifier and the ADC there shall be sufficient protection so that any voltage over the range will be clamped. The pathing condition is that there shall be two paths. The first path shall measure the voltage from 1 mV to 10 V. The second path shall measure the voltage from 10 V to 500 V.

Output from this stage should be exactly the same as the input of this stage except for the pathing and the clipping. The output of this stage shall be the now protected raw voltage which is divided into two paths: main path (1mV-10V) and a surge path (10V-500V). In general, no device shall handle current or voltage above what its range, clipping and incorporation shall be necessary.

Signal Conditioning

The input signal that is the protected and pathed telluric voltage shall be the input of this phase. The goal of signal conditioning is to filter the signal and attenuate/amplify it so that its range is within the range of the ADC. If the signal is a couple of millivolts and the ADC pins measure 0-3.3 volts, then a lot of resolution is missed out on. This section is the same in concept for both paths but slightly different specifications (voltage range and ADCs). The atomic tructure will first outline the main path then the transient path.

Main Path

At first, there shall be a signal attenuation by a factor of $8.\bar{3}$ so that the voltage out of the attenuator is a magnitude of ± 1.2 V. This attenuator shall match the impedance of both sides for minimal loss.

Connected to the attenuator shall be a low pass filter. 10 kHz is the requested sampling rate according to specification **TCS-MS2.1**. The filter shall not noticably impact the 0-10 kHz region but shall filter with a cutoff frequency of 15 kHz.

Connected to the filter shall be an amplifier. This amplifier will fit the now protected, pathed, attenuated, filtered signal to the range that the ADC expects. The ADC selected for the main path is the MCP3460R which has an input differential voltage range of $\pm V_{REF} \ / \ G$. The ADC has an intenal reference of 2.4V and a Gain which can be modified. If the gain is 1, then the ADC expects a differential voltage range of ± 2.4 V. This amplifier shall be galvanically isolated as another precaution against voltage surges.

The input to the main path shall be the protected and clamped telluric voltage. The output shall be the signal now attenuated by 8.333 filtered at a cutoff of 15kHz and amplified by a factor of 2 with galvanic isolation.

Surge Path

At first, there shall be a signal attenuation by a factor of 500 so that the voltage out of the attenuator is a magnitude of ± 1 V. This attenuator shall match the impedance of both sides for minimal loss.

Connected to the attenuator shall be a low pass filter. This low pass filter is to cut down on the aliasing when sampling. The filter cutoff frequency shall be 550 kHz (see the next section for rationale).

Connected to the filter shall be an amplifier. This amplifier will fit the now protected, pathed, attenuated, filtered signal to the range that the ADC expects. The ADC selected for the main path is the AD7352 which has an input differential voltage range of $\pm V_{REF}$. The ADC has an intenal reference of 2.048V. This amplifier shall have a gain of 2 and be galvanically isolated as another precaution against voltage surges.

The input to the surge path shall be the protected and clamped telluric voltage above. The output shall be the signal now attenuated by 500 filtered at a cutoff of 550 kHz and amplified by a factor of 2 with galvanic isolation.

These amplifiers shall be powered by the power subsystem referred there. Their power source will not be discussed to further extent in this subsection.

Analog to Digital Conversion

The final stage, analog to digital conversion, shall also have two paths. These ADCs shall be powered also by the power subsystem. Their power source will not be discussed to further extent in this subsection.

Main Path

The ADC selected for the main path is the MCP3460R. This ADC expects a signal within the range of ± 2.4 V of which the main paths signal is. The ouput of this ADC shall be the digital signal which as per secification **TCS-DS1.1** shall be within 10% of the true RMS of the raw voltage signal.

Surge Path

The ADC used for the surge path is AD7352 which has a maximum sampling rate of 3MS/s which is period of 333ns. This is much faster than any expected transient (including lightning). Instead, the signal shall be sampled at 100 MHz. By the Nyquist Criterion, the sample shall include harmonics up to 500 kHz. The harmonics at this frequency must remain accurate so that specification **TCS-DS1.2** is met. The output of the ADC shall be the digital signal which shall be within 20% of the true RMS of the raw voltage signal.

Summary

The DAQ subsystem shall have the following architecture. There shall be the input of the raw telluric voltages to the electrodes. This raw input shall be protected, that is clamped with the current limited. This signal shall be directed towards the main or surge path depending on the magnitude of the voltage. Then the signal shall be attenuated (by a factor of 8.333 and 500 respectively). After the attenuation, the signal shall be filtered by a low pass filter at frequency 15 kHz and 550 kHz respectively. These signals shall then both be amplified by a factor of 2. The output of the amplifiers shall both be galvaically isolated. The ADCs shall both sample the data and output a digital representation of the waveform with an accuracy of 10% or 20%, respectively, to the true RMS. This digital signal will be sent to the MCU.

4.

- Critique:

C4. Block Diagram Provided a block diagram of at least one solution?	Every subsystem shown. Power and signal paths labeled. System boundary, inputs, and outputs shown. Consistent with the text.	Complete. A few labels missing.	Main blocks shown. Interfaces unlabeled or inconsistent with text.	**Incomplete ("Power in → Coils → Car") or mixes physical and functional levels.**	No block diagram.	**2**
Instructor: The block diagram image is missing from the submission (see GTA comment). **GTA:** File not found						

- Text Before Revision:

an off-the-shelf USB power bank with isolated DC-DC and low-noise LDO post-regulation.

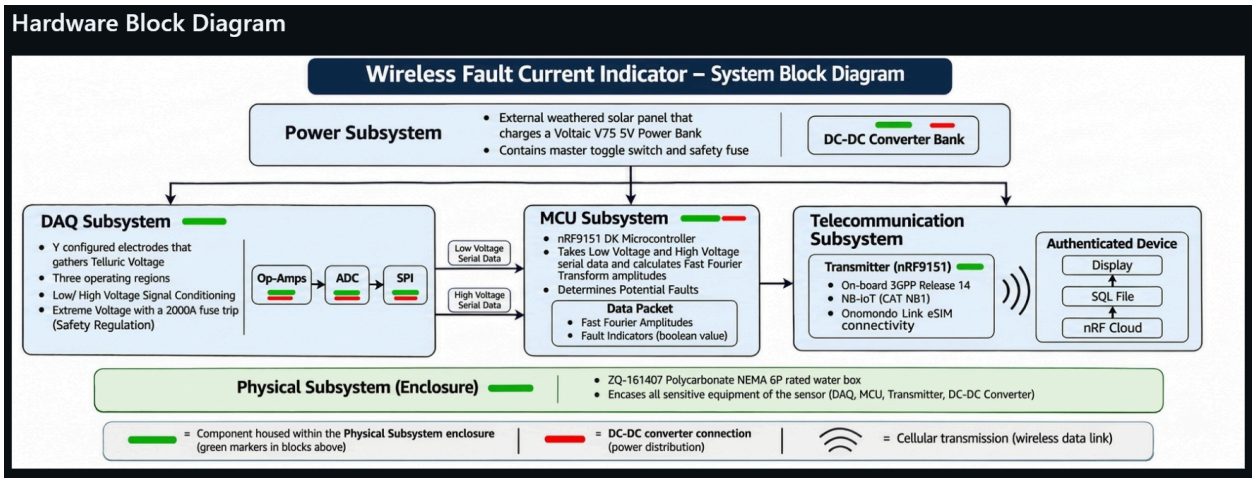
Hardware Block Diagram

Hardware Block Diagram

Operational Flow Chart

Operational Flow Chart

- Text Post Revision:

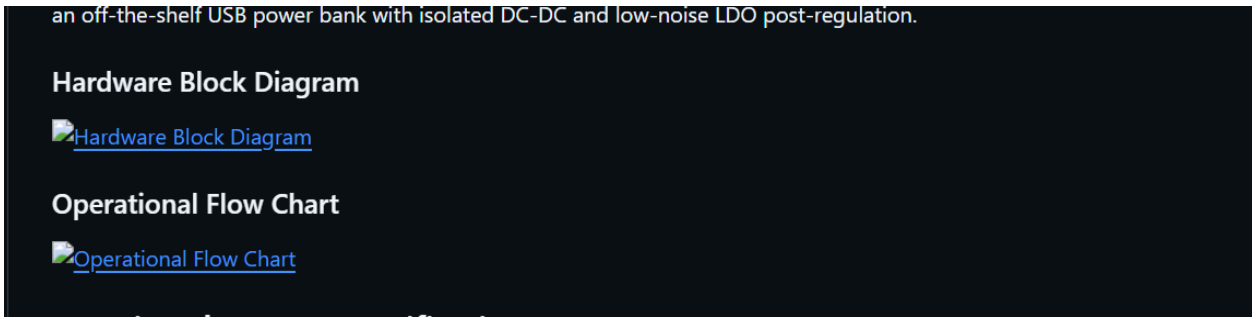


5.

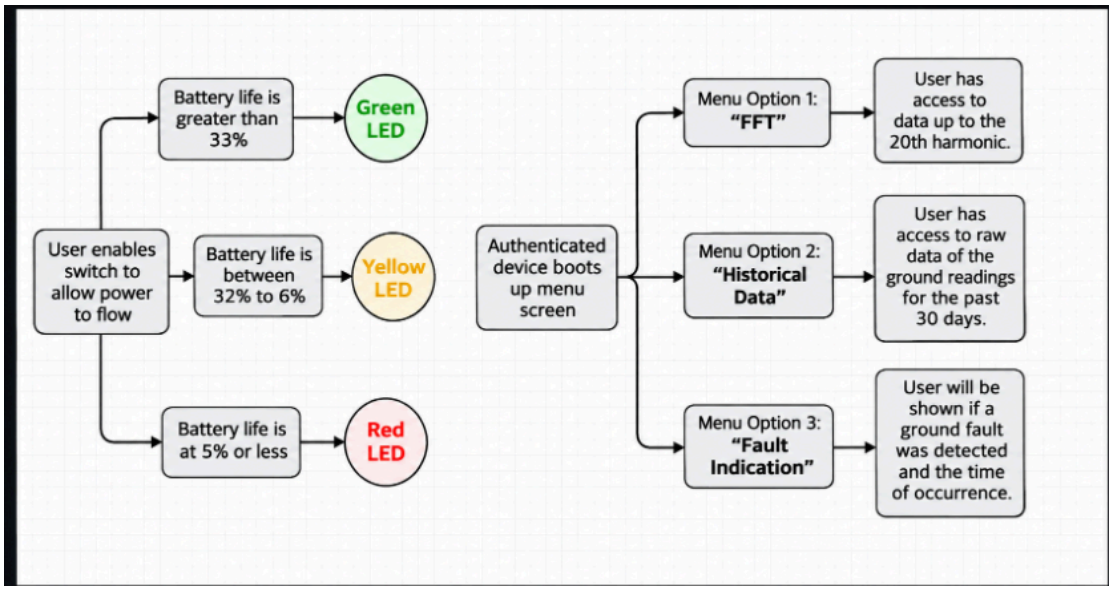
- Critique:

Requirement	5: Strongly Agree (Exceeds)	4: Agree (Achieves)	3: Neutral (Generally Meets)	2: Disagree (Below)	1: Strongly Disagree (Significantly Below)	Rating
C5. Flow Chart Created a flow chart of how at least one solution operates?	Standard symbols. Covers startup, normal operation, decision branches, fault handling, and shutdown.	Normal operation and main decisions. Fault handling partial.	Linear sequence only. No decisions or faults.	**Shows the project schedule instead of system operation, or unreadable.**	No flow chart.	**2**
Instructor: The flow chart image is missing from the submission (see GTA comment). **GTA:** File not found						

- Text Before Revision:



- Text Post Revision:



6.

- Critique:

C6. Materials, Resources, Steps Documented materials, resources, and implementation steps?	Preliminary BOM with quantities and costs. Lab equipment, software, and new skills listed. Sequenced steps with milestones matched to Capstone II.	BOM and steps present. Costs or equipment loosely estimated.	Parts listed generically. Steps vague.	**Partial list with no costs or sequence.**	Not addressed.	**2**
GTA: Missing for DAQ						

- Text Before Revision:

Prior Seal Team 2 didn't have a BOM for the DAQ. Additionally, the BOM was inconsistent and not dedicated to one section.

- Text Post Revision:

Seal Team 2's revision consisted of making a BOM for the DAQ. Each subsection has a listed BOM and cost is illustrated throughout the entire file.

7.

- Critique:

H2. Standards Identified governmental or industrial standards related to the concerns?	Standards cited by number with the relevant limit or requirement, mapped to each concern.	**Specific standards cited. Mapping to concerns partial.**	Standards body named without a document, or document does not apply.	"We will follow code."	Not addressed.	**4**
Instructor: Many standards are cited, but the document overstates or misapplies several, including NEC/NFPA 855 treatment of a USB power bank, IEEE 1547 applicability, and CAT IV assignment.						

- Text Before Revision:

NFPA 855	Stationary Energy Storage Safety	Battery Management & Storage	Pre-certified USB battery bank with thermal shutdown; firmware temperature monitoring.	Thermal cutoff active at ≥55 °C during charging and ≥60 °C during operation
IEEE Std 1547	DER Interconnection & Interoperability	Informational Grid Compatibility	Referenced strictly for utility grid voltage/frequency disturbance immunity limits (passive monitoring only; zero grid export).	Tolerates IEEE 1547 grid disturbance profiles without reset

- Text Post Revision:

NFPA 855 Scope Note	Stationary Energy Storage Safety (Scope Boundary)	Battery Storage	Clarifies that NFPA 855 applies to lithium-ion installations above the Table 1.3 threshold of 20 kWh; a 72.36 Wh portable pack relies on UL 2056 / UL 1642 certifications.	Compliance with manufacturer UL certifications

2. NFPA 855 Scope Boundary and Lithium Safety

- **Scope Clarification:** NFPA 855 applies to lithium-ion energy storage installations exceeding the Table 1.3 threshold of 20 kWh per fire area (Section 1.3) [SS-2]. Because the a power pack is a portable, small-scale storage device (72.36 Wh), formal NFPA 855 stationary compliance does not directly apply.
- **Integration Strategy:** Safety and thermal protection rely directly on the power bank's built-in UL certifications (UL 2056 [SS-27] / UL 1642 [SS-28]) and hardware BMS protection.

IEEE was referenced purely for informational utility grid disturbance immunity limits (passive monitoring only). Because the Telluric Current Ground Fault Indicator (TC-GFI) is a non-contact, passive monitoring device that does not export power back into the utility grid, there is no need to implement active smart-inverter grid-support functions or formal utility interconnection compliance. (This standard was removed).

8.

- Critique:

H3. Standards Application and Verification Applied identified standards to specific design decisions and described how compliance will be verified?	Several standards applied with clause-level detail to quantified requirements; at least one verified by test or analysis.	Several standards applied to specific requirements or design choices, with credible compliance verification planned.	**At least one standard applied to a specific requirement or design choice; verification limited or generic.**	Standards cited but not meaningfully applied to the design.	No standards identified or applied.	**3**
Instructor: The standards table maps some standards to choices and target metrics, but clause-level support, applicability justification, and credible verification procedures are inconsistent.						

- Text Before Revision:
  - **This was changed throughout the whole project. Seal Team 2 recommends the instructor and graduate teaching assistant to re-read Relevant Standards and Constraints and Measures of Success. However, a brief overview of prior text will be provided below:**
    The original document identified engineering standards and made those connections within the subsystems. However, the applicability of standards and the procedures for verifying compliance were not explained in a clear, consistent manner.

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Text Post Revision: The document now provides a more detailed connection between engineering standards, design decisions, and measures of success.

9.

- Critique:

H4. Broader Factors Explained how global, cultural, social, environmental, or economic factors influenced the solution choice?	Two or more factors visibly changed the selection, with evidence in the decision matrix.	**One factor clearly influenced the choice.**	**Factors discussed but not linked to the selection.**	Generic statement ("solar is good for the environment").	Not addressed.	**3.5**
Instructor: Economic, rural access, environmental, cybersecurity, and social factors are discussed, but most are not visibly used in selection matrices.						

- Text Before Revision:

Ethical, Professional, and Standards Considerations

Ethics:

- Honest Performance claims: Report detection limits, sensitivity, and failure modes accuratley, not just the best case results. Overstating reliablity on a safety device is a ethical issue.
- Fail-Safes: If the power is lost, the battery degrades, or a component fails. The sensor should indicate its own failure rather than producing flase or weak reading.
- Codes of Ethics: Cite IEEE Code of Ethics, and the NSPE Code of Ethics.
- Data and Privacy: Logs that transmit across shared grid data should state who owns it and how its protected.
- Intellectual Property and Attributions: Give credit to prior work and individuals whom helped fairly.
- Data Integrity vs. Alarm Fatigue: Sensor data must accurately decouple true earth fault conditions. Misclassifying a telluric wave as noise risks missing genuine dangerous ground fault currents.
- Affordability and Safety: Reducing cost should not come at the expense of essential safety and reliability. Balance affordability with dependable measurments.

Data acquisition electrode constraints: Since testing will be performed near high-tension lines, there are important safety considerations to be investigated. For one, there is a chance of pertubing underground utilties. To avoid this, OSHA 29 CFR 1926.651(b) requires that the utility owner is contacted before any excavating occurs, and Tennessee's Underground Utility Damage Prevention Act (TCA 65-31-101) requires everyone who digs to contact Tennessee 811 at least three (3) working days' notice before starting a project [SG1], [SG2]. The testing location should also be verified on what is permitted within the right-of-way, as per TVA's Transmission Right of Way team [SG3]. To also prevent electric shock or similar injuries during installation, it is recommended by OSHA 1910.333(c)(6) to minimize the hazard through implementation of work practice such as handling techniques and insulation [SG4]. Beause of this, personal protective equipment and proper insulation should be diligently inspected, with these ratings given by 29 CFR 1926.97 and OSHA 29 CFR 1926.957, respectively [SG5], [SG6].

Telecommunication Subsystem Constraints

Independently designed transmitter constraints: All wireless transmitters must adhere to the authorization constraints enumerated in the FCC 47 CFR section 2.911 document. In particular, any independently constructed RF transmitter/Receiver network in need of authorization "shall be submitted in writing to a Telecommunication Certification Body (TCB) in a manner prescribed by the TCB." [4]. Other sections of Part 2 of FCC 47 CFR, notably, 2.1033, 2.1046, and 2.1057 must be considered when designing a transmitter that meets the federal frequency and bandwidth regulations. These legal constraints on transmitter design makes it favorable to choose an already approved (commercial) device for the transmitter.

Integrated commercial transmitter/modem constraints: When integrating a commercial transmitter into a communication module, the design of the communication module should adhere to the *FCC KDB 996369 D01: Transmitter Module Equipment Authorization Guide* [5]. As a consequence, if the design of the transmitter of the Ground Fault Sensor incorporates a commercial device, there are still legal requirements that need to be considered.

Use of Public Mobile Services Constraints When designing a transmitter to a cellular network, *FCC CFR 47 Part 22* provides limits for frequencies and effective radiated power. Section 22.352 also goes into interference protection protocols as well [6]. This constraint limits the available bandwidth and power output of the transmitter of the Ground Fault Sensor in the case of using a cellular communication architecture.

Microcontroller Subsystem Constraints

Since the microcontroller will be directly wired to the data acquisition unit, it would be regulated under Supbpart B of Title 47 of the FCC. Under Title 47 of the FCC, the microcontroller would be considered a Class A digital device, which is a device used for industrial and commercial use. Since it would also emits radio frequencies as a byproduct of the function of its clock mechanism, the microcontroller would be considered an unintentional radiator. Because of this, it is regulated as an unintentional radiator, it requires authorization such as the Supplier's Declaration of Conformity to ensure it abides the FCC before it can be marketed [7].

Effective December 23, 2027, there will be a procurement ban on products using semiconductors sourced from blacklisted entities due to Section 5949 of the James M. Inhofe National Defense Authorization Act for Fiscal Year 2023 [8]. This could be a negligible constraint since it would only affect the market for U.S. federal government.

There are additional regulations pertaining to ensuring cybersecurity of the components being connected to the power grid. The North American Electric Reliability Corporation, also known as NERC, requires that security procedures are implemented for the reliable operation and security of the Bulk Electrical System. Under Section B 1.2.5, it requires that any software used for the MCU is verified to comply with strict cyber security standards to ensure power grid safety [9].

Since the microcontroller is apart of a larger fault detection apparatus, it is required by the IEEE Guide for Testing Faulted Circuit Indicators (495-2025) in Section 4 to having an operating condition of -40&degC to 85&degC [10].

IEEE 1613 pertains to the testing and environmental requirements of communication devices with electric power apparatus applications. for this system deals with fault detection, the microcontroller will experience voltage surges. To increase the practicality of the device, meeting Class 1 standards in IEEE 1613 allows the device to experience temporary error as long as the device has a self-recovery system [9].

There is also IEEE 1613.1 that is related to transmission and distribution applications. In this document, it defines Zone B as facility with equipment outside of an enclosed area. For our purposes, the MCU would need to be prepared for Zone B use, meaning that the environmental standards due to being outside of a boundary or fenced are [11].

IEC-61010-1 pertains to category ratings for safety requirements of equipment used for measurement, among other uses. Since the MCU will be used for monitoring power systems, it would need to be a CAT IV rating, which requires transient protection level up to 12 kilovolts [12]. Because of this requirement, protection of the microcontroller will be paramount.

Power System Constraints

Voltage Regulation: The power distribution and regualtion stages must maintain a tight DC voltage tolerances with minimal ripple to prevent high frequency noise[13].

Transient and Surge Handling: The system must absorb rapid load steps and isolate the system from external voltage spikes or lightning surges via proper supercapacitor buffering[14].

Electrical and Isolation: The analog and processing unit must maintain galvanic isolation to prevent high Ground Potential Rise and faults from damaging downstream circuitry or creating ground loops [15], [16].

Power Budget: Desinged for remote, unattended deployment the systems power must be minimized to ensure the hybrid storage can sustain operation[3].

Financial: The total Bill of Materials needs to reamin with limits while using reliable components.

Legal Constraints

Regulatory and Safety Codes: All internal wiring, power distribution, and grounding must strictly comply with NEC (NFPA 70) requirements, specifically Article 250 (Grounding and Bonding), Article 690 (Solar PV Systems), Article 706 (Energy Storage Systems), and Article 725 (Low-Voltage Control Circuits) [SS-1].

Thermal and Chemical Limitations: Battery charging and discharging parameters must respect chemical limits (preventing lithium or thermal runaway under NFPA 855 guidelines) [SS-2].

Geographical and Site Restrictions: Field deployment requires correct physical orientation and spacing of sensing probes relative to local telluric vectors to help with meaningful data collection.

Applicable Standards and Design Implementation

To ensure safe operation, signal integrity, and regulatory compliance across environments ranging from field sites to high-voltage substations, the power subsystem incorporates design criteria from the following key engineering codes and standards:

Standard / Code	Title / Focus Area	Applicable Subsystem / Aspect	Design Integration and Incorporation Strategy	Target Verification Metric
NEC (NFPA 70) Art. 250	Grounding & Bonding	Power Distribution & Chassis	Single-point reference grounding; metallic chassis bonded to earth reference.	Chassis ground resistance under $0.1, \Omega$
NEC (NFPA 70) Art. 690	Solar PV Systems	Auxiliary Solar / MPPT Stage	Overcurrent protection inline with PV feed; rated wire sizing and terminal isolation.	Fuse and wire sizing matches NEC ampacity limits
NEC (NFPA 70) Art. 706	Energy Storage Systems	Battery & Supercap Storage	Fused battery connections, polarity protection, and physical isolation within enclosure.	Fuse clearing and safe battery disconnect verified
NEC (NFPA 70) Art. 725	Class 1, 2, & 3 Remote-Control / Power-Limited	Low-Voltage Power Rails	Physical segregation of low-voltage DC signals (3.3V / 5V) from high-voltage interfaces.	Minimum $6.3\ \text{mm}$ PCB trace clearance

NFPA 855	Stationary Energy Storage Safety	Battery Management & Storage	Pre-certified USB battery bank with thermal shutdown; firmware temperature monitoring.	Thermal cutoff active at ≥55 °C during charging and ≥60 °C during operation
IEEE Std 80	Safety in AC Substation Grounding	Enclosure & Ground Probes	Galvanic isolation barrier prevents Substation Ground Potential Rise (GPR) from damaging microcontrollers.	Withstands 2 $\mathrm{kV_{RMS}}$ Hi-Pot test for 60 seconds
IEEE Std 4	High-Voltage Testing Techniques	Input Power Protection	Transient Voltage Suppressor (TVS) diodes and isolation barrier designed for surge withstand.	Withstands 3 kV transient impulse surge
IEEE Std 1100	Powering & Grounding Sensitive Electronics	Analog Front-End (AFE) Rails	Low-noise LDO filtering and isolated DC-DC converter prevent noise coupling into sensitive telluric band.	Noise floor under $10, \mu\mathrm{V_{RMS}}$ ($0.01\ \mathrm{Hz} - 100\ \mathrm{Hz}$)
IEEE Std 1547	DER Interconnection & Interoperability	Substation Control Interfacing	Defines isolation and power quality bounds when interfacing with utility grid buses.	Fault-free operation during grid disturbances
IEEE Std C37.91	Protective Relaying for Transformers	Substation Sensor Placement	Structural shielding and transient clamping to withstand transformer magnetic fields and fault currents.	Uninterrupted function during high EMF exposure
USB 2.0 / USB-C PD Spec	USB Power & Delivery Standard	Primary Field Input (VBUS)	Standardized 5 V DC intake interface using commercial USB power banks and wall adapters.	Voltage stable within $4.75\mathrm{V} - 5.25\mathrm{V}$ under load

Resources

Budget

- Text Post Revision:

Ethical, Professional, and Standards Considerations

Cultural Impacts

Cultural impacts would include the rise in understanding through-the-soil electrical engineering applications. This field is one that is explored by researchers including Dr. Charles Van Neste, the customer of this project. If this project proves successful, there is a possibility that more students and researchers will take an interest in this field of study. Another potential impact would be for new interest it DIY-style devices with power system applications to arise. Since this project is composed of undergraduate students, a successful outcome would possibly result in other undergraduate students, potentially high school students, to attempt to design new devices with power system applications based on areas of personal intrigue. A more negative impact that could occur is desecration of lands deemed sacred by indigenous populations. This is a risk due to the easier implementation of the TC-GFI in rural and remote locations. To mitigate this affect, implementations will abide by government guidelines related to historically significant places. Initial implementations will occur in Cookeville, Tennessee, so this will not be a concern at the current moment. If testing will later be conducted within land controlled by a Native American Reversation, discussions with the tribal government will need to be conducted.

Societal Impacts

Societal changes that this Wireless Fault Current Indicator could bring is that if a device that is a inexpensive solution to fault current detection can bring more use cases to less developed areas of the world. For example, electrical co-operatives serve relatively larger areas of land compared to more urban and developed municipals. Because of this, it requires the deployment of a fleet of work vehicles to locate a fault when it occurs. If the product can have a minimized price, this can increase the number of units to be purchased by the interested parties, meaning that a greater area can be served. If a potential triangulation feature could be implemented, it can be deduced that this device would simplify the process of locating ground faults in a power system. Applications in developing countries with less robust power infrastructure are also available due to the minimally invasive installation process and straight-forward operation in addition to the cost of the device. Urban and developed areas would not receive the main benefits of this device due to greater difficulty of installation due to the installation process requiring soil embedment. For mitigation of this technological disadvantage, further research could be conducted once the concept can be demonstrated to work in remote and rural areas.

Environmental Impacts

One environmental consideration would be the intrusion of the soil. The electrodes will need to be dug into the ground, perturbing the natural environment during the process. Since the amount of digging performed will be regulated by other regulations such as Tennessee 811 [SGB] and OSHA 29 CFR 1926.651(b) [SGA], this environmental impact will be mitigated to a minimum.

Public Health Impacts

Since this device will be used for power grid applications, it should not have ease of access to the public. In addition, this device will also be designed to be autonomous after configuration, meaning that perpetual proximity of individuals need not be required. Because of this, there is minimal public health concerns with this project.

Public Safety Impacts

One important consideration for public safety is clear indication of its use cases. Since testing of this device will be coordinated in the vicinity of a substation, it is a wise decision to clearly mark that it is the property of the university, as having an unmarked electronic device in a box near the electric could be perceived as a potential threat to public safety. In addition, since this device is going to monitor voltages, proper safety measures will be established to ensure injury does not occur during the installation of the electrodes or the testing of equipment. Relevant standards will be OSHA 1910.333(c)(6), OSHA 29 CFR 1926.97 and OSHA 29 CFR 1926.957, which will be discussed further in a later section [SGD], [SDE], [SDF].

Economic Impacts

Digging into the soil for the electrodes can also pose economic risks if unverified excavation damages underground utilities. Relevant standards will be OSHA 29 CFR 1926.651(b) and Tennessee's Underground Utility Damage Prevention Act (TCA 65-31-101), which will be discussed further in the following section [SGA], [SGB]. Since this device is intended to reduce the area for locating a ground fault, this could reduce the manpower needed to locate a sustained ground fault when it occurs, saving time and money. However, battery replacement would be a routine maintenance requirement, meaning that a worker will have to visit the site. The best way to reduce the cost of the battery as well as visiting the site on a routine basis is maximizing battery life to reduce the amount of visits to the installation site per year.

Ethical considerations:

- **Code of Ethics**: Our group will abide by the codes of ethics given by the IEEE Code of Ethics and NSPE Code of Ethics, ensuring that we will perform our moral and professional responsibilities as future engineers.
- **Testing Responsiblities**: The testing location should also be verified on what is permitted within the right-of-way, as per TVA's Transmission Right of Way team [SGC]. There will also be safety considerations related to OSHA regulations discussed in later sections.
- **Honest Performance**: Report all performance results regardless if they fail to meet specifications. Report detection limits, sensitivity, and failure modes precisely. Misrepresentating reliability is to be considered a violation of our codes of ethics.
- **Affordability and Safety**: The ideal outcome is to create something that is affordable, safe, and reliable. However, safety and reliability are paramount should not be adulterated to increase affordability. Affordability should be balanced with dependable measurements
- **Data and Privacy**: Logs that transmit across shared grid data should state who owns it and how its protected.
- **Intellectual Property and Attributions**: Proper attribution of curators of prior work and individuals who assisted in the creat90j results of this project.
- **Fail-Safes**: If the power is lost, the battery degrades, or a component fails. The sensor should indicate its own failure rather than producing false or weak reading.
- **Data Integrity vs. Alarm Fatigue**: Sensor data must accurately decouple true earth fault conditions. Misclassifying a telluric wave as noise risks missing genuine dangerous ground fault currents.

Relevant Standards and Constraints

Given all of these considerations, the following includes relevant standards, regulations, and constraints for our system that will dictate the design and testing of our device:

Title of Standard/Code	Testing (T) or Design (D) Aspect	Requirements	Implementation
FCC 47 CFR Section 2.911 [4]	D	Any independently constructed RF transmitter/Receiver network in need of authorization "shall be submitted in writing to a Telecommunication Certification Body (TCB) in a manner prescribed by the TCB	Choose an pre-approved (commercial) device for the transmitter
IEEE Std 4	D	Dictate how components and isolation barriers must be engineered to handle transient overvoltages	Transient Voltage Suppressor (TVS) diodes and isolation barrier designed for surge withstand, withstanding 3 kV transient impulse surge
IEEE Std 80	D	Defines the level of voltage the MCU must be able to withstand	Galvanic isolation barrier prevents Substation Ground Potential Rise (GPR) from damaging microcontrollers, withstanding 2 kV_{RMS} Hi-Pot test for 60 seconds
IEEE 495-2025 [10]	D	Operating condition of -40°C to 85°C	Select an MCU that meets the operating ratings required by this standard
IEEE Std 1100	D	Sensitive electronic hardware requires clean, conditioned power rails free from high-frequency switching harmonics and broad-band noise that can mix into analog baseband channels	Low-noise LDO filtering and isolated DC-DC converter prevent noise coupling into sensitive telluric band, with noise floor under $10, \mu V_{RMS}$ $(0.01\ \text{Hz} - 100\ \text{Hz})$
IEEE Std 1613 [9]	D	Given the nature of fault detection, the device shall have a self-recovery system as defined by a Class 1 device with communication abillities used in a power system context	Use of watchdog timers and ensuring no corrupted data is transmitted through data rollback

IEEE Std 1613.1 [11]	D	Meet stricter environmental standards defined by Zone B due to being outside a boundary or fenced area	Select an MCU that meets strict environmental standards and weatherproof it
IEEE Std C37.91	D	Details the protection schemes (such as harmonic-restrained differential protection 87T and backup overcurrent 50 / 51) required to safeguard power transformers	Structural shielding and transient clamping to withstand transformer magnetic fields and fault currents, with uninterrupted function during high EMF exposure
NEC (NFPA 70) Art. 250 [SS-1]	D	Mandates maximum resistance to ground	Single-point reference grounding; metallic chassis bonded to earth reference, chassis ground resistance under $0.1, \Omega$
NEC (NFPA 70) Art. 690 [SS-1]	D	Dictate design choices for the direct current photovoltaic circuits for safety and sizing requirements	Overcurrent protection inline with PV feed; rated wire sizing and terminal isolation, and fuse and wire sizing matches NEC ampacity limits
NEC (NFPA 70) Art. 706 [SS-1]	D	Requires a verified safe battery disconnect, overcurrent protection devices, prevention of reverse-voltages, and prevention of accidental contact of the DC terminals and busbars	Fused battery connections, polarity protection, and physical isolation within enclosure, with fuse clearing and safe battery disconnect verified
NEC (NFPA 70) Art. 725 [SS-1]	D	Regulates the spatial relationship between Class 2/3 low-voltage circuits and higher voltage power conductors	Physical segregation of low-voltage DC signals (3.3V / 5V) from high-voltage interfaces, with minimum 6.3 mm PCB trace clearance
NERC Section B 1.2.5	D	Any software used for the MCU is verified to comply with strict cyber security standards to ensure power grid safety	Utilize Nordic's nRF Cloud and TLS libraries to secure the data

OSHA 1910.333(c)(6) [SGD]	T	Required to minimize the hazard of electric shock or similar injuries through implementation of work practice such as handling techniques and insulation	Refer to OSHA 29 CFR 1926.97 [SGE] and OSHA 29 CFR 1926.957 [SGF] for ratings of personal protective equipment and insulation, respectively
OSHA 29 CFR 1926.651(b) [SGA]	T	Requires that the utility owner is contacted before any excavating occurs	Contact any nearby utilities to unsure that no utility could be damaged
Tennessee's Underground Utility Damage Prevention Act (TCA 65-31-101) [SGB]	T	Requires everyone who digs to contact Tennessee 811 at least three (3) working days' notice before starting a project	Ensure that the group will contact Tennessee 811 before any digging occurs

The ethics portion was re-written to incorporate discussions of impacts, positive and negative, across multiple sectors of the word, in addition to further discussions for standards and potential implementations.

10.

- Critique:

H5. Mitigating Constraints Described additional constraints to mitigate the concerns?	Each concern converted to a measurable constraint added to the specification list.	**Constraints stated. Some not measurable.**	**Mitigation stated as intention ("we will make it safe").**	Constraints unrelated to the stated concerns.	Not addressed.	**3.5**
Instructor: Some concerns become constraints, but many mitigations remain qualitative and are not added to a unified measurable specification list.						

- Text Before Revision:

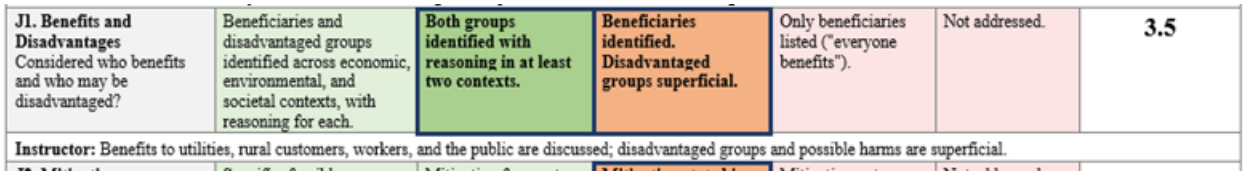
Same text as pre-revision for #9

- Text Post Revision:

Same text as post-revision for #9

11.

- Critique:



J1. Benefits and Disadvantages Considered who benefits and who may be disadvantaged?	Beneficiaries and disadvantaged groups identified across economic, environmental, and societal contexts, with reasoning for each.	**Both groups identified with reasoning in at least two contexts.**	**Beneficiaries identified. Disadvantaged groups superficial.**	Only beneficiaries listed ("everyone benefits").	Not addressed.	**3.5**
Instructor: Benefits to utilities, rural customers, workers, and the public are discussed; disadvantaged groups and possible harms are superficial.						

- Text Before Revision:

Same text as pre-revision for #9

- Text Post Revision:

Same text as post-revision for #9

The ethics portion was re-written to incorporate discussions of impacts, positive and negative, across multiple sectors of the word, in addition to further discussions for standards and potential implementations.

12.

- Critique:

J2. Mitigation Explicitly stated how they will mitigate the impact of the solution?	Specific, feasible mitigation for each disadvantaged group, built into the project or recommended to the sponsor.	Mitigation for most groups.	**Mitigation stated in general terms.**	Mitigation not connected to the groups identified.	Not addressed.	**3**
Instructor: Mitigations are not explicitly tied to disadvantaged groups or negative impacts.						

- Text Before Revision:

Same text as pre-revision for #9

- Text Post Revision:

Same text as post-revision for #10

The ethics portion was re-written to incorporate discussions of impacts, positive and negative, across multiple sectors of the word, in addition to further discussions for standards and potential implementations.

Line by Line Critiques

13.

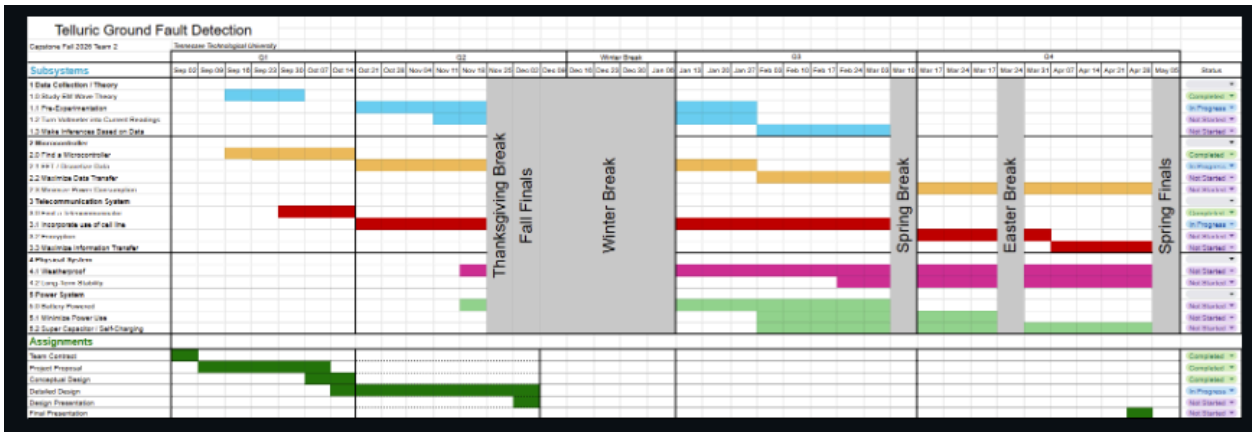
- Critique:

- (P5) Line 448: Missing Gantt chart. A file path for image is given, but the actual file isn't there

- Text Before Revision:



- Text Post Revision:



14.

- Critique:

- (C2) Lines 694 and 711: LoRaWAN is claimed to have excessive power consumption but is also listed as meeting all specifications and constraints for power consumption under all relevant conditions in the comparison table.

Discussion was provided clarifying why we chose Cellular over LoRaWAN. See Comparison of Communication Media subsection of the Comparative Analysis section of Team 2's conceptual design.

15.

- Critique:

- (C2/C6) Lines 726, 746, and 750: The nRF9160 itself is a compact surface mount SiP that could be extremely difficult to integrate, as it requires a somewhat complex array of external components in order to function (see Nordic's hardware integration guide for the part). Based on the wording used in this section, it sounds like you intend to use a development kit or breakout board for the nRF9160. If this is the case, it should be explicitly mentioned that you are talking about the development board and not the discrete SiP module.

We have clarified throughout our document that we will be using the nRF9151 DK.

16.

- Critique:

- (C2/C6) Lines 764-774: Comparison of solutions and a selection of proposed solution are missing for the power subsystem.

- Text Before Revision: Text was mislabeled/placed in incorrect spot

- Text Post Revision: Moved and appropriately labeled for ease of sight

17.

- Critique:

- (C1, C2, C3, C6): DAQ subsystem is never addressed. What sort of signals will be read? What sort of conditioning/filtering/amplification will be needed? What will be used to convert the data to a format usable by the MCU?

- Text Before Revision:

N/A

- Text Post Revision:

Atomic Subsystems, Comparative Analysis, Measures of Success all incorporate these changes.

DAQ Protection and Pathing

The DAQ shall be protected from surges over 500 V and overcurrents which is the primary go of the protection phase. Each node shall have a resistor on it which sufficiently prevents overcurrents (10k). The first path shall clamp at (or close to) 10 V so that the path is saturated, not opened. Same will be for any voltage above 500 V and the second path. These clamps sha be capable of conducting within 30μs, the standard period of a lightning strike surge. The selected device shall be a metal oxide varistor (MOV) that will respond by clamping over 500V within a couple of microseconds (<5uS). The resistors responsible for suppressing the current shall be rated high enough to withstand transients within the response time of the MOV or sustained high voltage (hundreds of volts) without varying greatly in value over time.

The rest of the circuit shall also be protected if there is leftover voltage after the initial protection stage. Before the amplifier and the ADC there shall be sufficient protection so that any voltage over the range will be clamped. The pathing condition is that there shall be two paths. The first path shall measure the voltage from 1 mV to 10 V. The second path shall measure the voltage from 10 V to 500 V.

Output from this stage should be exactly the same as the input of this stage except for the pathing and the clipping. The output of this stage shall be the now protected raw voltage which is divided into two paths: main path (1mV-10V) and a surge path (10V-500V). In genera no device shall handle current or voltage above what its range, clipping and incorporation sha be necessary. As a result, each of the following stages shall be protected from overvoltage by grounding diode or some other simple but precise protection scheme.

18.

- Critique:

P4. Stakeholder Specifications Listed specifications and measures of success?	Every specification is quantitative with units, tolerance, and a verification method, and traces to a stakeholder need.	Quantitative with units. Traceability or verification missing for a few.	**Mix of quantitative and qualitative terms ("fast", "efficient").**	Mostly qualitative ("high efficiency, safe, low cost"). No measures of success.	No specifications listed.	**3**
Instructor: Many specifications are vague or untestable: "minimized," "weatherproof," "all outdoor conditions," "abnormalities," and "efficiently." The 24-hour power target conflicts with battery replacement >1 month.						

- Text Before Revision: Autonomous Uptime: Achieve continuous 24-hour operation under load.
- Text Post Revision: ****Autonomous Uptime:**** Achieve continuous 24-hour operation under load. - ****Long-Term Field Operations:**** Support extended unattended field deployments for up to 3 months.